



Exploring the potential of explainable AI in brain tumor detection and classification: a systematic review

Lincy Annet Abraham¹ · Gopinath Palanisamy¹ · Goutham Veerapu¹ · J. S. Nisha¹

Received: 20 June 2025 / Accepted: 23 September 2025 / Published online: 18 November 2025
© The Author(s) 2025

Abstract

The analysis and treatment of brain tumors are among the most difficult medical conditions. Brain tumors must be detected accurately and promptly to improve patient outcomes and plan effective treatments. Recently used advanced technologies such as artificial intelligence (AI) and machine learning (ML) have increased interest in applying AI to detect brain tumors. However, concerns have emerged regarding the reliability and transparency of AI models in medical settings, as their decision-making processes are often opaque and difficult to interpret. This research is unique in its focus on explainability in AI-based brain tumor detection, prioritizing confidence, safety, and clinical adoption over mere accuracy. It gives a thorough overview of XAI methodologies, problems, and uses, linking scientific advances to the needs of real-world healthcare. XAI is a sub-section of artificial intelligence that seeks to solve this problem by offering understandable and straightforward and providing explanations for the choices made by AI representations. Applications such as healthcare, where the interpretability of AI models is essential for guaranteeing patient safety and fostering confidence between medical professionals and AI systems, have seen the introduction of XAI-based procedures. This paper reviews recent advancements in XAI-based brain tumor detection, focusing on methods that provide justifications for AI model predictions. The study highlights the advantages of XAI in improving patient outcomes and supporting medical decision-making. The findings reveal that ResNet 18 performed better, with 94% training accuracy, 96.86% testing accuracy, low loss (0.012), and a rapid time (~ 6 s). ResNet 50 was a little slower (~ 13 s) but stable, with 92.86% test accuracy. DenseNet121 (Adam W) achieved the highest accuracy at 97.71%, but it was not consistent across all optimizers. ViT-GRU also got 97% accuracy with very little loss (0.008), although it took a long time to compute (around 49 s). On the other hand, VGG models (around 94% test accuracy) and MobileNetV2 (loss up to 6.024) were less reliable, even though they trained faster. Additionally, it explores various opportunities, challenges, and clinical applications. Based on these findings, this research offers a comprehensive analysis of XAI-based brain tumor detection and encourages further investigation in specific areas.

Keywords Convolution neural networks(CNNs) · Artificial intelligence (AI) · Explainable AI (XAI) · Machine learning (ML) · Gradient-weighted class activation mapping (Grad-CAM) · Local interpretable model-agnostic explanations (LIME) · SHapley additive explanations (SHAP)

Abbreviations

AI	Artificial intelligence
BCM-CNN	Brain classification model–CNN
BraTS	Brain tumor segmentation challenge
BR35H-CNN	Brain tumor classification with 35-layer hierarchical CNN
CAD	Computer-aided diagnosis
CAM	Class activation mapping
CGAN	Conditional generative adversarial network
CNN	Convolutional neural network
CT	Computed tomography
DDI	Deep dream image
DL	Deep learning
FCN	Fully convolutional network
FLAIR	Fluid-attenuated inversion recovery
GAN	Generative adversarial network
GBM	Glioblastoma multiforme
GRU	Gated recurrent unit
Grad-CAM	Gradient-weighted class activation mapping
KBT	Kaggle brain tumor
KNN	K-nearest neighbors
LIME	Local interpretable model-agnostic explanations
LRP	Layer-wise relevance propagation
MICCAI	Medical image computing and computer-assisted intervention
MID	Medical image dataset
ML	Machine learning
MRI	Magnetic resonance imaging
OASIS	Open access series of imaging studies
PET	Positron emission tomography
PRISMA	Preferred reporting items for systematic reviews and meta-analyses
REMBRANDT	REpository for Molecular BRain Neoplasia DaTa
RF	Random forest
RNN	Recurrent neural network
SHAP	SHapley additive explanations
SVM	Support vector machine
TCIA	The cancer imaging archive
VAE	Variational autoencoder
VGG16	Visual geometry group 16-layer CNN
ViT	Vision transformer
WOA	Whale optimization algorithm
XAI	Explainable artificial intelligence
YOLO	You only look once

1 Introduction

Brain tumors are among the most challenging conditions to diagnose and treat because of their complexity and the wide range of symptoms that patients may experience. Although it has been demonstrated that artificial intelligence improves the accuracy and usefulness of brain tumor discovery, questions have been raised regarding the safety and dependability of AI algorithms in the health field due to their complexity and difficulty in understanding. To address this challenge, XAI has emerged, providing succinct and comprehensible clarifications for the choices made by AI models. The medical community and the AI system can work together more effectively and with greater accountability when clinicians can comprehend the underlying causes of the AI's predictions thanks to XAI techniques. This study summarizes the latest advancements in XAI-based brain tumor detection, including different methods for producing explanations, and talks about the opportunities and difficulties of applying XAI in clinical settings. Similar to other body organs, the brain can develop issues, including growths called brain tumors (Yurtsever et al. 2024). These tumors develop from abnormal cell growth in various regions of the brain, potentially leading to severe neurological complications and a significant decline in the patient's quality of life. Most people concur that brain tumors rank among the deadliest diseases worldwide, and they significantly impact death rates across all age groups (Gyamfi et al. 2022). In terms of medical diagnosis and treatment, this condition presents difficulties. These tumors can appear as pituitary, glioma, meningioma, or in cases where there is no tumor at all.

Due to the composite structure of the brain, the nearly 120 different types of tumors that have been discovered so far are inspiring to identify due to their varied features and sizes. Edema, necrotic centers, and tumorous tissues are examples of tumorous areas that must be precisely identified and segmented to aid in diagnosis and action planning. It is unbearable to overstate the status of automatically identifying and categorizing medical images, especially when diagnosing brain tumors (Ahmed et al. 2024; Raghavendra et al. 2023). For brain tumors to be treated more rapidly and give patients better outcomes, they must be detected and categorized early. This is followed by the same MID. It is the conventional method of finding brain tumors. PET, CT, and MRI scans are examples of medical imaging methods that are essential for identifying brain tumors. The most famous system for examining the brain's structures is MRI. Because MRI is extensively used and real in investigating brain structures, it was selected as the preferred modality for our investigation study. Although MRI is capable of modeling regions of interest, the differences in strength levels brought about by different machine settings make it difficult to differentiate cancers (Abdusalomov et al. 2023). To this end, DL approaches are creating tools for brain cancer classification and segmentation. They provide precise and reliable information about the kind, location, and existence of tumors in computer-aided brain tumor diagnosis systems.

PRISMA is a set of rules that makes sure systematic reviews and meta-analyses are clear, complete, and can be repeated. It combines a 27-item checklist and a flow diagram to concisely explain study selection, methods, and results. The PRISMA framework is frequently mandated by journals and funding agencies, and it has seen widespread adoption in healthcare and artificial intelligence in medicine. Its main goals are to increase the quality of reviews while decreasing bias.

The Fig. 1 represents a comprehensive search was conducted across multiple reputable databases to capture a wide array of potentially relevant studies. The databases searched

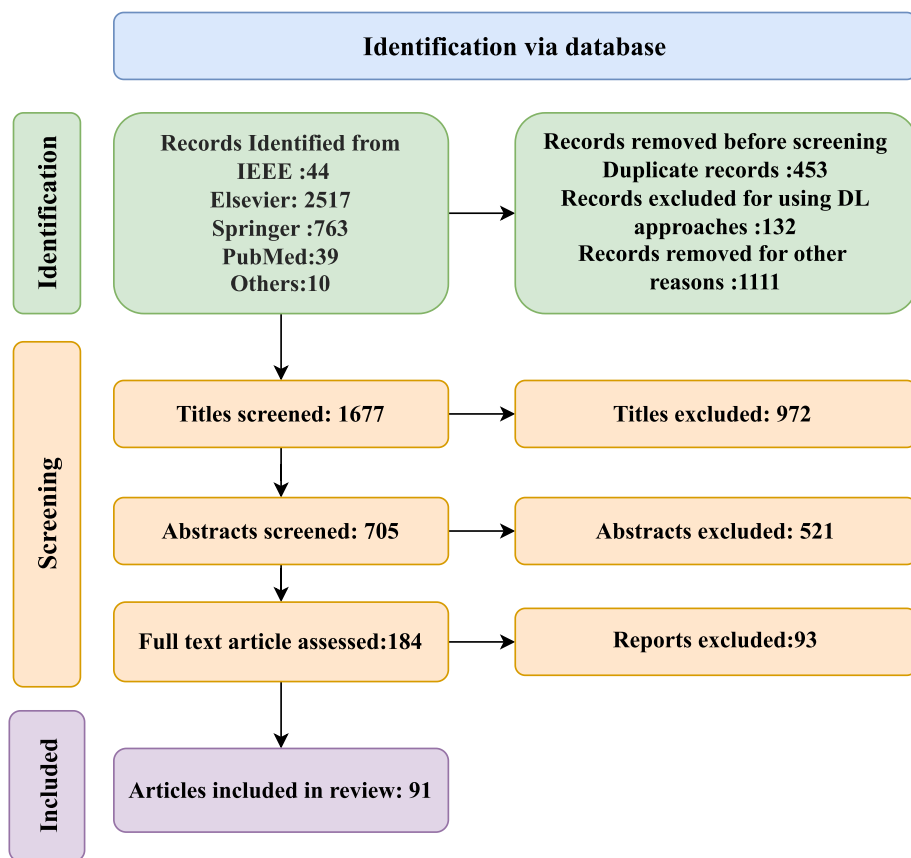


Fig. 1 PRISMA chart showing the research strategy for proposed systematic review

included IEEE (44 records), Elsevier (2517 records), Springer (763 records), PubMed (39 records), and others (10 records), resulting in a total initial pool of studies. Before the screening phase, 1,696 records were excluded as they consisted primarily of books or materials deemed irrelevant to the review scope. This left 1,677 studies to undergo title screening, where each title was carefully evaluated for relevance to the research question. From this, 972 records were removed due to irrelevance.

The abstracts of the remaining 705 studies were then assessed in detail. During this abstract screening phase, 521 articles were excluded based on several reasons: 212 were inaccessible despite efforts to retrieve them, 300 were considered out of scope for the specific focus of the review, and 9 were published in languages not eligible for inclusion.

Following abstract screening, 184 full-text articles were reviewed thoroughly to judge their eligibility against pre-established inclusion criteria. During this full-text assessment, 93 articles were excluded due to not meeting the criteria, such as inadequate methodology or lack of focus on XAI and brain tumor detection.

Ultimately, 91 high-quality and relevant studies were included in the final systematic review. This careful selection process, using clearly defined criteria and multiple screening

stages, strengthens the validity of the review by ensuring that only pertinent and methodologically sound research contributes to the analysis.

The novelty of this review lies in the integration of real-time brain tumor detection and classification with XAI, providing not only accurate predictions but also interpretable explanations for clinical decision-making. The key contributions of this paper are as follows:

- *Development of a robust detection–classification pipeline* The proposed method integrates YOLO for tumor detection with advanced classifiers for precise tumor type identification (e.g., glioma, meningioma, pituitary tumor).
- *Incorporation of explainable AI* Provides heatmaps and attention maps that explain why and where the model focuses, ensuring decisions are interpretable for radiologists and clinicians.
- *Improved diagnostic accuracy* Achieves high precision and recall compared to conventional CNN or transformer-based models without explainability.
- *Trustworthy decision support system* Helps clinicians validate AI predictions, reducing the risk of misdiagnosis and enhancing acceptance of AI in medical practice.
- *Comprehensive evaluation* Tested on benchmark brain tumor MRI datasets with comparative analysis against state-of-the-art methods.

2 Background

There are two categories of deep learning models, input-dependent explanations and input-independent explanations. Directly explain the model without requiring network input to identify and elucidate the features associated with the significant influence of the prediction results. The segmentation and classifications are directly clarified by the use of medical imaging (Ali et al. 2020; Kumar and Vinod Kumar 2024). Typically, perturbation-based and gradient-based approaches use instance-level explanations to explain these tasks (Rustom et al. 2023). The number of input features that are computed to the various resources required to achieve better results determines the running time for both methods, which are used in the early stages to observe changes in output under various inputs. The next step is to decide between accuracy and interpretability. Fei Yan (2023) provides an estimate of the input feature's importance in their description of GBM. Accuracy and computational loss are not balanced in the explainability training process. In recent years the use of the extension method followed by CAM. Grad-CAM and Guided Grad-CAM are used to improve this procedure (Aeman et al. 2023).

Researchers have employed target detection networks, followed by deep learning (DL), such as U-Net, VGG, ResNet, and GoogLeNet in medical submissions. It is central to analyzing the series of diseases. Tian et al., used FCN for tumor cataloging and origination to improve the tumor localization effect (Rehman et al. 2020). The proposed model ACAE is used to classify the different shares and quickly detect the suspected areas. In recent years, Research scholars have used XAI to assess the model results. 3D CNN classification and explainability used to detect Alzheimer's disease Murthy et al. (2022) Rosnelly et al. (2023). XAI is used for Natural image tasks followed by insufficient tasks to medical field. Explainability of Medical images has a significant role in the analysis of the great demand

and scalability followed by the degrees of health professionals followed by model results and operations.

Figure 2 The dataset contains a collection of brain MRI pictures from Kaggle annotated to indicate checks the brain tumor is present or not. The datasets vary from the different brain tumors. Data preprocessing as MRI images help to remove the noise-normalized intensity values and provide the segment through brain regions from various backgrounds. The MRI brain scans are measured to find the types of brain tumors, this also comprises the various types of images such as T1-weighted, T2-weighted and FLAIR sequences which is also mentioned in the segmentation masks. The data type of concern method is 3D medical imaging data. The labels of this data set contain the tumor segmentation. Based on this research used the BR35H -CNN model with XAI technique, the dataset set contains 3060 brain images related to MRI in both tumor and non-tumor cases (Balwant 2022). The core research of this dataset is to classify the classification among brain tumor types, explainable AI model development followed by to interpret the classification result. Image Preprocessing means providing Standardized imaging protocols, minimizing relics, and lowering inter-subject variability are some of the challenges we face when preprocessing primary MRI data. The process of feature extraction itself is difficult; to capture significant tumor characteristics, careful selection is needed. Data split contains the testing set, validation set, and training set. The classification model consists of DL and ML. Generating model consists of two types of methods such as LIME and SHAP.

The workflow diagram in Fig. 3 consists of six steps data collection, data preparation, data preprocessing, feature extraction, training and validation, and model evaluations. Data preparation means collecting the samples in image format followed by the various acquisition parameters. Data preprocessing is used to check the quality of DL analysis. Feature extraction, which means fine-tuning and identifying the brain tumors from MRI scans. Training and validation, which means evaluating the performance using various metrics

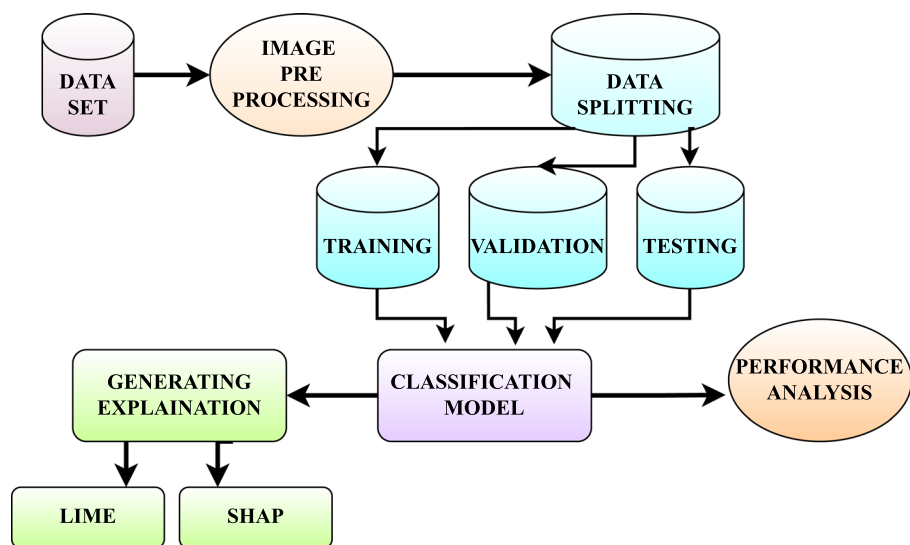


Fig. 2 Flow diagram of brain tumor classification and detection

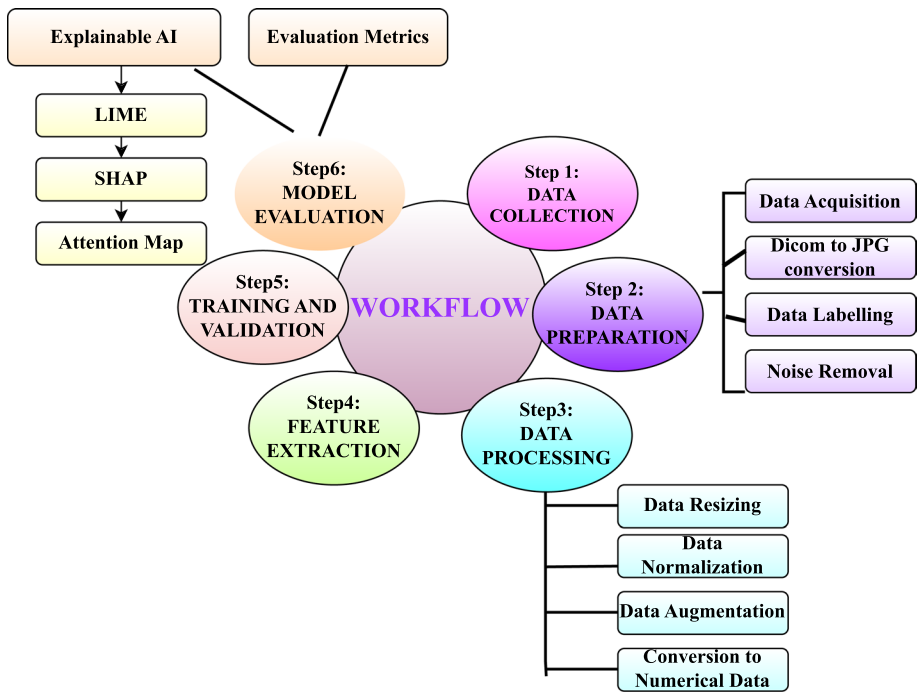


Fig. 3 Work flow diagram of various steps used in the research approach

including accuracy, precision, recall, and F1-score. The workflow diagram obtained the highest average level of accuracy with positive instances.

3 Algorithms for detection/classification of brain tumor

AI technique combines ML and DL with revolutionized medical imaging. These approaches are enabled through automated extraction of relevant features with medical descriptions such as MRI and CT Scans. These all help to improve the accuracy and detection of brain tumors. To help radiologists, make better clinical decisions, AI models are made to recognize patterns in information that do not directly affect people's judgment.

3.1 Machine learning algorithms

As a branch of AI and ML, it entails training models on a dataset to identify trends and generate predictions. The following machine learning procedures have been used to notice brain tumors, Brain tumors are classified using SVM, which uses features taken from imaging data. One collaborative learning method for tasks involving regression and classification is RF. A straightforward and efficient classification technique that has been used to identify brain tumors is KNN (Maqsood et al. 2022). These algorithms rely on manually extracted tumor area characteristics like texture, shape, and intensity. Decision trees operate by dividing data according to naturally interpretable feature values. It is possible to visualize

each decision node, demonstrating how characteristics such as tumor size, shape, or texture influence classification choices. Decision trees can be used as a variety of features, such as texture and intensity, or they can be used to determine the spatial relationship between MRI scans to classify brain tumors as benign. It is important to make the decision tree's explainable AI components naturally interpretable so that feature importance analysis can be readily extracted. The decision path is easier to understand when the different tools, such as TreeSHAP are visualized. This algorithm's benefits include transparency and ease of visualization.

3.2 Deep learning algorithms

Among DL techniques, CNNs have occurred as the most hopeful approach to brain tumor detection. The need for physical feature extraction can be avoided by using CNNs to mechanically learn hierarchical features from raw medical images. Deep learning's capacity to manage sizable datasets and extract complex patterns straight from image pixels is what makes it so effective at detecting brain tumors (Rao et al. 2024). Gliomas, meningiomas, and pituitary tumors are among the brain tumors that CNN-based models are very good at identifying and categorizing (Vankdothu and Hameed 2022). CNNs are a key module of contemporary diagnostic AI because of their capacity to generalize across various kinds of brain tumor datasets. Pre-trained models, such as VGG16 and ResNet, are commonly optimized for brain tumor classification tasks to minimize the need for massive amounts of labeled data in transfer learning (Montaha et al. 2022).

The Fig. 4 illustrates the trade-off between complexity and explainability in ML techniques, categorizing them into traditional ML and DL. Explainability is represented by the x-axis, which rises from left to right, and complexity by the y-axis, which rises from bottom to top. Although they offer clear decision-making processes, traditional machine

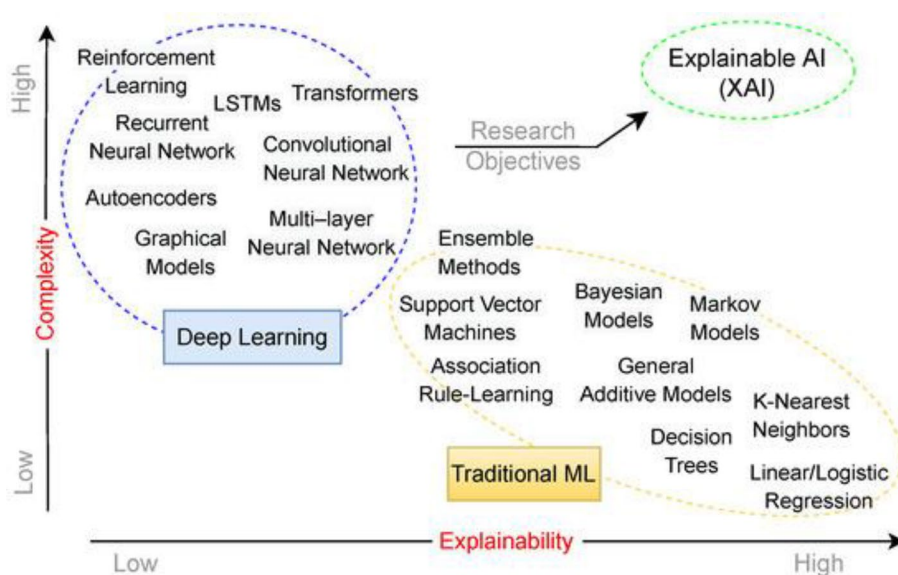


Fig. 4 Complexity and explainability between ML and DL

learning techniques like Bayesian models, decision trees, and linear/logistic regression are positioned on the lower complexity and higher explainability side. However, they may have trouble handling intricate data patterns. Neural networks, CNNs, RNNs, and transformers are examples of deep learning models that fall on the side of greater complexity and lesser explainability. Despite being excellent at identifying complex relationships in data, these models are frequently viewed as black boxes since they are difficult to interpret. The graphic also emphasises XAI research goal, which is to close the gap by making complicated AI models easier to understand without sacrificing their predictive accuracy. Making deep learning models easier to understand is still a major obstacle to developing reliable and open AI systems as the technology develops.

Deep learning architectures interact differently with explainability tools, influencing the nature and utility of generated explanations. CNNs, with their hierarchical spatial feature maps, are well-suited for gradient-based approaches such as Grad-CAM, which highlight discriminative regions in medical images. Transformer-based models, on the other hand, inherently incorporate self-attention mechanisms, enabling attention visualization and attribution methods (e.g., attention rollout, SHAP) to reveal feature dependencies across image regions. Generative models, including VAEs and GANs, can be integrated with counterfactual explanations to synthesize plausible alternative cases, thereby providing insight into feature-level reasoning and decision boundaries. Together, these interactions demonstrate that the choice of XAI technique should be closely aligned with the underlying model architecture to maximize both fidelity and clinical usability.

Transfer learning plays a crucial role in brain tumor classification by leveraging pre-trained models on large datasets to improve accuracy and reduce training time on limited medical data. Key points include:

- *Improves accuracy* Transfer learning uses features learned from large, diverse datasets (often non-medical) and fine-tunes them with brain tumor MRI data. This significantly improves classification accuracy for tumors such as benign vs malignant or specific tumor types like glioma, meningioma, and pituitary tumors, often reaching accuracy above 98–99%.
- *Reduces data requirements* By building on pre-trained deep neural networks (e.g., ResNet, VGG, MobileNet), transfer learning reduces the need for large labeled medical image datasets, which are costly and difficult to obtain.
- *Enhances generalization* Fine-tuning pre-trained models on diverse brain tumor datasets helps models generalize better to unseen samples and different imaging conditions.
- *Speeds up training* Transfer learning accelerates model convergence, enabling faster development and deployment of AI diagnostic tools.
- *Supports multi-class classification* Transfer learning enables effective differentiation among multiple tumor categories (e.g., glioma, meningioma, pituitary, no tumor) with high precision and recall.

In summary, transfer learning significantly boosts the performance, efficiency, and scalability of AI systems for brain tumor classification, making it a vital technique in modern medical image analysis. Transfer learning plays a vital role in brain tumor classification by leveraging pre-trained deep learning models to improve accuracy, reduce the need for extensive labeled medical datasets, and accelerate training times. It allows models to uti-

lize learned features from large, diverse datasets and fine-tune them for brain tumor MRI images, achieving high precision and recall in distinguishing tumor types such as glioma, meningioma, pituitary tumors, and normal conditions. Transfer learning enhances generalization to unseen cases and supports multi-class classification, making it a powerful technique for efficient and effective brain tumor diagnosis.

3.3 Explainable AI (XAI)

Explainable AI in brain tumor detection and classification aims to make the decision-making process of complex AI models transparent and interpretable while maintaining high accuracy. Traditional deep learning models, though powerful in identifying tumors from MRI images, often operate as black boxes with limited explainability, making it difficult for clinicians to trust and validate their predictions. XAI techniques, such as Grad-CAM, SHAP, and LIME, are integrated with CNNs to highlight the most relevant tumor regions and critical features in the images, providing visual and quantitative explanations for the model's decisions.

By focusing on abnormal brain tissues responsible for different tumor subtypes, XAI-enhanced models improve classification performance and generalizability, achieving accuracy as high as 99% on known data and about 95% on unseen datasets. This combination of interpretability and accuracy helps clinicians better understand, trust, and adopt AI tools for diagnosis. Furthermore, XAI methods validate that the models do not rely on irrelevant features or overfit the data, making them robust and reliable for real-world clinical applications. The use of XAI in this domain represents a significant advancement toward transparent, accountable, and effective medical AI systems for brain tumor diagnosis and treatment support.

XAI techniques such as feature importance methods, SHAP, LIME, rule extraction, and visualization approaches like activation maps—break down these complex models into understandable components. They reveal which input features have the most influence, explain individual predictions locally, and highlight the critical regions or factors contributing to decisions. This transparency fosters trust, accountability, and facilitates identification of biases or errors that would otherwise go unnoticed in black-box systems.

Table 1 provides a comparative summary of widely used XAI techniques in brain tumor detection. Grad-CAM and CAM remain the most common methods due to their intuitive visual heatmaps that highlight tumor regions, which are particularly valuable in MRI-based localization tasks. However, these methods are often sensitive to noise or restricted by architectural requirements. In contrast, model-agnostic approaches such as SHAP and LIME provide feature-level attributions with stronger theoretical grounding, though they are computationally expensive and sometimes unstable. The table highlights the trade-offs between interpretability, computational cost, and clinical applicability, underscoring the need for careful selection of XAI methods depending on the intended diagnostic workflow.

As summarized in Table 2, each XAI method exhibits different levels of fidelity, robustness, and interpretability, highlighting the inherent trade-offs when applied to clinical decision support.

While XAI methods have shown promise in improving model transparency, several limitations remain. Perturbation-based techniques such as LIME and SHAP often struggle with stability and reproducibility when applied to medical images, as small perturbations in high-

Table 1 Summary of common XAI methods, their advantages, limitations, and typical use in brain tumor detection

XAI method	Advantages	Limitations	Typical use
Grad-CAM	Visual heatmaps, intuitive for clinicians	Sensitive to noise, may miss global decision logic	MRI tumor localization
SHAP	Model-agnostic, strong theoretical basis, consistent feature attributions	Computationally expensive, less suited for real-time use	Feature attribution and model interpretability
LIME	Simple, flexible, works with any classifier	Local explanations only, unstable for small perturbations	Explaining classifier decisions on medical images
CAM	Straightforward visualization of discriminative regions	Requires specific network architecture (global average pooling)	Weakly-supervised tumor localization

Table 2 Comparison of widely used XAI techniques in terms of fidelity, robustness, clinician usability, and trade-offs

XAI method	Fidelity	Robustness	Clinician usability	Trade-offs
Grad-CAM	High (localized heatmaps)	Moderate (sensitive to architecture changes)	High (visual, intuitive)	May overlook global feature interactions
LIME	Moderate (local approximation)	Low–Moderate (unstable explanations)	Moderate (feature weights need interpretation)	Computationally expensive; less consistent
SHAP	High (theoretically grounded attributions)	High (consistent across models)	Moderate (requires tailored visualization)	Complex and slow for large models
Integrated gradients	High (satisfies sensitivity axioms)	Moderate (depends on baseline choice)	Moderate (numeric attributions less intuitive)	Explanations less visually interpretable

dimensional data can lead to inconsistent explanations. These methods also assume feature independence, which is rarely the case in complex medical imaging scenarios. Moreover, their computational overhead makes them impractical for real-time clinical decision support. Gradient-based methods like Grad-CAM, although visually intuitive, may overlook global feature interactions and provide explanations that are architecture-dependent. These limitations highlight the need for more reliable, efficient, and clinically validated XAI approaches before large-scale deployment in healthcare settings.

The main research gaps in XAI for brain tumor diagnosis and classification are centered around the following key challenges:

- *Limited interpretability in complex models* Many deep learning approaches in brain tumor classification use complex architectures that make them difficult to interpret. This lack of interpretability can hinder clinicians' understanding of the model's decision process and can sometimes cause the models to focus on irrelevant or normal features, reducing reliability and clinical acceptance.
- *Balancing model complexity and transparency* Simplifying AI models can enhance transparency and deployment efficiency in clinical settings, but may compromise feature extraction and accuracy. Complex models risk overfitting to irrelevant features; simpler models may fail to capture nuanced tumor characteristics.
- *Generalizability and dataset diversity* Current XAI-based models often suffer from limited generalizability due to small or homogeneous datasets. Expanding the datasets with more varied MRI, and potentially integrating other modalities like PET or CT, would improve model robustness across a broader patient population.
- *Clinical relevance of explanations* Existing XAI techniques (such as Grad-CAM, SHAP, LIME) help provide interpretable results, but there is still a gap in offering clinically meaningful explanations that are easily trusted and actionable by medical professionals. More advanced XAI methods tailored to medical imaging are needed to make explanations directly relevant to clinical decision-making.
- *Multimodal and real-time integration* Most current models rely mainly on MRI images, often neglecting other important data sources like clinical history, genetics, and histological data. Integration of multimodal information and development of real-time, interpretable systems for hospital workflows are emerging as important research directions.
- *Model uncertainty and user interaction* Present XAI methods often lack the ability to communicate the uncertainty of predictions or provide interactive, customized explanations needed for clinical risk assessment and user trust.

Explainable AI addresses the black-box limitations of AI models by making their decision-making processes more transparent, understandable, and interpretable for humans. Here's how XAI tackles these challenges:

- *Feature importance* XAI techniques highlight which input features most influenced the model's decision, helping users understand what factors contributed to a specific prediction.
- *Local and global explanations* Methods like LIME and SHAP provide clear, human-readable explanations for individual predictions (local) or for the model's overall behavior (global). This helps users see why a model arrived at a certain output in specific cases as well as in general patterns.
- *Visualization tools* Techniques such as activation maps and attention mechanisms visually reveal what parts of an input (such as regions in a medical image) the model focuses on, offering insight into the model's reasoning.
- *Rule extraction and surrogate models* XAI can use simpler, interpretable models (like decision trees or rule-based systems) to approximate and explain the behavior of complex black-box models in understandable terms.
- *Bias and error detection* By making AI decisions transparent, XAI enables the identification of model biases and errors, supporting regulatory compliance and increasing user trust in critical fields such as healthcare and finance.

- *Building trust and enabling accountability* XAI allows users, regulators, and stakeholders to trace, validate, and audit how and why a model makes its decisions, which is essential for responsible deployment in high-stakes situations.

Furthermore, XAI enhances regulatory compliance and supports clinical adoption in sensitive domains like healthcare by enabling meaningful human oversight and validation of AI decisions. Although challenges remain in balancing the trade-off between interpretability and model complexity, XAI represents a crucial advancement in transforming black-box AI into explainable, trustworthy systems.

3.3.1 Literature review

The Table 3 summarizes recent research on the application of XAI in brain tumor detection and classification. The investigations encompass several strategies, including hybrid deep learning models, ensemble techniques, and counterfactual explanations. The main benefits of these research are that they make models easier to understand, boost trust and clinical relevance, and give visual explanations for how decisions are made. Nonetheless, obstacles persist, including substantial computing expenses, the necessity for extensive datasets, and the intricacy of models. Lack of specific implementation instructions is another reason why certain findings can't be applied clinically or generalized to other types of tumors. Despite these problems, XAI techniques offer great potential in tackling interpretability issues in AI models, making them more helpful and trustworthy in clinical environments.

3.4 Medical imaging modalities for brain tumor detection

The following medical imaging modalities have primarily employed AI techniques to detect brain tumors: MRI is the most popular graphical method for brain tumor detection. They offer high-resolution pictures and can differentiate between various brain tissue types. MRI scans can be used for AI models that can detect growth patterns that could be signs of a tumor (Ullah et al. 2022). CT scans can still be used to find brain tumors, despite being less detailed than MRIs. AI algorithms can improve CT image analysis, particularly when MRI scans are not available in an emergency. Early tumor detection can be aided by PET scans, which determine metabolic activity in the brain. AI can help interpret PET data to improve tumor detection (Singh et al. 2020).

3.5 Challenges in AI-based brain tumor detection

Despite the significant progress artificial intelligence has made in detecting brain tumors, several problems still need to be fixed,

3.5.1 Data quality and availability

AI model performance is strongly influenced by the caliber and volume of training data. Annotated datasets of high quality are frequently scarce, particularly in the field of medical imaging (Kodipalli et al. 2024). The primary issue with medical imaging is that hospital protocols differ. Many scanning parameters, such as resolution, thickness, contrast level, and

Table 3 Summary of selected studies on explainable AI in brain tumor detection

References	Key focus	Methodology	Merits	Demerits
Houssein et al. (2025)	A systematic review of XAI in the processing of medical images	Assessments the use of XAI for the purpose of AI openness	Emphasizes the necessity for interpretability in clinical AI	There is no new model presented, simply a review
Nahiduzzaman et al. (2025)	XAI for finding brain tumors	Brings together deep learning and XAI methods	Increases trust and gives visual explanations	Limited details on how to use it in the clinic
Lamba et al. (2025)	A hybrid model that combines ResNet101, Xception, and LIME	Improves the accuracy and ease of understanding of classifications	Gives a detailed explanation for clinical validation	Requires a lot of computation
Gundogan (2025)	Using XAI with hybrid deep learning for multi-class classification	Utilizes XAI for the categorization of brain tumors from MRI	Results that are easy to understand and very accurate	Computational overhead
Farhan et al. (2025)	Grad-CAM++ is an ensemble-based deep learning method	Uses MobileNetV2, DenseNet121, and Grad-CAM++ for tumor detection	Strong performance that is useful in the clinic	Requires large dataset for training
Chen et al. (2025)	Counterfactual Explanations for Tumors (TACE)	Utilizes counterfactuals for the change of tumor-specific characteristics	Increases the rate of successful classification	Only certain types of tumors can be treated
Shawon et al. (2025)	Cost-sensitive neural networks using XAI for detection	Deep neural networks that are cost-sensitive and use XAI approaches	Fixes data imbalance and promises correctness	Making models more complicated
Li et al. (2024)	MProtoNet for MRI data with several parameters	Interpretable model based on cases without human-annotated labels	Improves measures for understanding	Needs a lot of info and comments

orientations, are included in this. The impact of this challenge has prompted an analysis of the variation among the images used to train the AI model, which failed to match the training data when the real-world data was deployed. A standardized imaging protocol is used to address this issue and lessen its impact. The distribution of the various classes in the dataset may not be balanced, which is the issue with data imbalance.

Brain tumor datasets frequently have an under representation of some tumor types or a disproportionate number of healthy samples relative to tumor samples. Two trained AI models that are probably biased have the effect of making it difficult to detect underrepresented classes, which lowers the sensitivity. Data augmentation, under-sampling, and over-sampling are some of the methods used to solve this issue. A combination of technological advancements, moral behavior, and teamwork are needed to address issues with data availability and quality in AI-based brain tumor detection. Although AI has a lot of promise to improve the detection of brain tumors, the development of trustworthy, generalizable, and objective models depends on the availability of diverse, high-quality, and well-annotated data. Several of these issues can be resolved with developments in data augmentation, federated learning, transfer learning, and multimodal integration, opening the door for more precise and easily available AI-based medical devices.

3.5.2 Generalization across datasets

Due to differences in patient populations and scanning protocols, AI models trained on one dataset might not simplify well to data from other organizations or imaging equipment. The term generalization across datasets describes an AI model's capacity to function effectively on fresh, untested data, even if the data originates from a different source, distribution, or context than the training data. It is the key component for providing guarantee and robustness among AI models for detecting brain tumor detection in healthcare industries. After analysis, obtaining a good generalization based on AI-based brain tumor detection is difficult. These types of difficulties have differed from the various constraints such as patient information, image condition, and dataset. Here, the various types of problems in generalization included inter-patient variability and imbalance across the dataset. The major problem in AI-based brain tumor detection is generalized followed by the different types of datasets. Various types of strategies are used as transfer learning, data augmentation, domain adaption, multi-model learning, these are used to improve the model's robustness to overcome these challenges. Here to develop the AI model through various clinical settings and patient populations will also require the creation of representative and diverse datasets, consistent annotations, and the application of strategies like active learning.

Several publicly available datasets as shown in Fig. 5 are widely used for brain tumor detection, offering diverse imaging modalities, annotations, and tumor types. The BraTS dataset, provided by MICCAI, is one of the most popular and contains multimodal MRI scans (T1, T1c, T2, and FLAIR) with annotated tumor subregions, making it ideal for deep learning applications. The Figshare Brain Tumor Dataset consists of MRI images categorized into tumor and non-tumor classes, useful for basic classification tasks. Another notable dataset is REMBRANDT, hosted by TCIA, which includes MRI scans of glioma, glioblastoma, and astrocytoma cases. Additionally, TCIA provides multiple brain tumor datasets, including glioblastoma and meningioma cases, with detailed clinical and imaging metadata. The Harvard Whole Brain Atlas and OASIS offer MRI scans covering both brain tumors and neurodegenerative conditions. Kaggle also hosts various brain tumor datasets, such as MRI images labeled by tumor type, making them accessible for machine learning research. These datasets enable researchers to develop and refine AI models for tumor detection, segmentation, and classification, ultimately contributing to advancements in medical diagnostics.

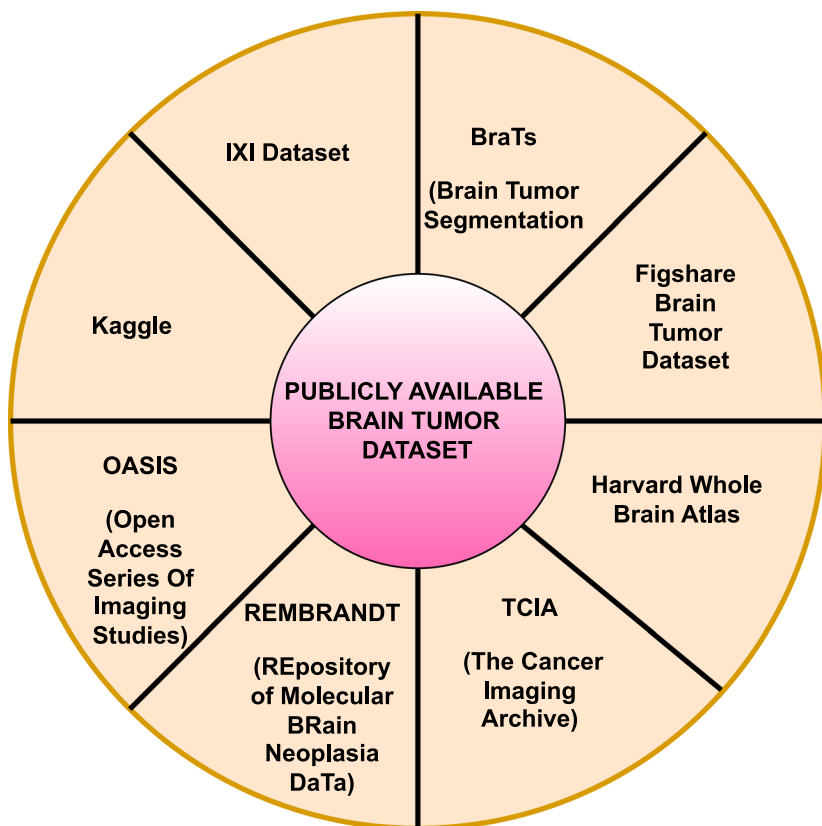


Fig. 5 Datasets for Brain tumor detection

3.5.3 Interpretability

AI models are accomplished by attaining high accuracy, but they are regularly regarded as black boxes, making it inspiring to understand the reason for predictions. This restricts their use in clinical settings, where interpretability and transparency are essential. A major challenge in AI-based brain tumor detection is generalization across datasets. AI models are covered by Black boxes, which contain high accuracy, which is the reason for predicting the results. Various types of restrictions are used for the clinical setting, which helps to understand the predictions. Here discussed the various challenges in AI-based Brain tumor detection among datasets. If the AI model was generalized should be affected by the different types of imaging protocols followed by the various types of constraints. Different techniques are used as domain adaptation, data augmentation, multi-model learning, and regularizations, these all help to increase the performance of a given model, and it helps easily overcome these types of problems. To develop the AI model was generalized along with clinical areas with patent information. Two types of strategies are used to check the modality difference and also overcome the various types of problems. The result is consistent after the interpretations. To explain the various types of complex models used by some classification tools such as SHAP and LIME. After interpretation analysis, the various ranges of patient

information should require the development of fairness awareness among the correction techniques. To overcome these types of issues AI model also improved the clinical settings with the adoption and trust of brain tumor detections.

Future work for improving XAI interpretability is focused on several innovative approaches and research directions:

- *XAI by design* Moving beyond post-hoc explanations, researchers advocate building inherently explainable models. This involves designing new AI architectures that embed interpretability into their structure, making transparency a foundational feature rather than an add-on.
- *Adaptive and user specific explanations* Tailoring explanations to users' expertise levels and specific needs, whether clinicians, patients, or regulators, by using interactive visualizations and non-technical language. This makes AI systems more accessible and usable across different stakeholders.
- *Causal and counterfactual explanations* Developing techniques that not only tell what influenced a model's decision but also answer why and how potential changes in input would alter outcomes. This brings deeper insights into the model's causal reasoning.
- *Neuro-symbolic and hybrid models* Integrating neural networks with symbolic reasoning to combine high performance and human-readable explanations—a trend that has shown promise in making even deep learning models interpretable.
- *Interactive and real-time explanations* Allowing end-users to interact with AI systems for personalized, on-demand explanations and making AI reasoning transparent during operational decision-making.
- *Federated and privacy preserving explainability* Enabling XAI in decentralized and sensitive settings (such as healthcare) without revealing private or proprietary data—solving a key barrier in high-stakes domains.
- *Democratization and tooling* Expanding cloud-based XAI toolkits and APIs, making advanced interpretability features widely accessible and easy to implement.
- *Domain-specific interpretability* Customizing interpretability approaches for domains like healthcare, finance, and law, where regulatory requirements and decision impacts are significant.

These directions will foster greater trust, accountability, and adoption for AI solutions by making models not only accurate but truly interpretable and transparent in real-world scenarios.

3.5.4 Regulatory approval

Provides the guarantee through clinical practice efficiency, AI-based tools used for validation, and follows the regular approval procedures. From that stage, we followed the regular framework to check equity, transparency, and clinical efficiency. To describe the various problems related to post-market monitoring and to check the interpretability among the models. The requirement is based on checking the data privacy issue and clinical validations. To overcome these issues, AI developers are to approve the process and increase the dependability of the system used in a clinical environment. To implement the various practices related to data security and privacy.

3.5.5 Complexity of diverse data

The multi-model datasets are used to provide the information through the brain such as “DWI, FLAIR, T1, T2, and Fat sat images”. These types of datasets are difficult to integrate and also have limited accessibility. MRI Data used for to analysis as in Fig. 6 of the main MRI imaging was inefficient. To develop the algorithms related to the MRI dataset, which means overcoming these issues.

3.5.6 Class imbalance and tumor rarity

Developing AI brain tumor detection systems faced various issues such as class imbalance and tumor rarity. Here addressed some issues in the Machine learning model as accuracy and sensitivity. To examine issues in brain tumor detection as class imbalance and tumor rarity. Large datasets contain the more normal MRI scans, AI model is used to profit the majority of the class and ignore the limited class results of imbalance. In terms of tumor detection, to mention the model sensitivity and accuracy. These are all the issues faced for reduced sensitivity, overfitting to the class, and false negatives. To overcome these issues followed by some methods such as Resampling techniques, class weighted loss function and anomaly detection and transfer learning programs. The uncommon issue of brain tumors as a rarity occurred in a small percentage of MRI Scans. Here, various misclassifications as lack of training data. Here the solution of this problem used a combination of various techniques including synthetic data generation, data augmentation, and few-shot learning. Cutting-edge machine learning techniques are used to identify the classification of uncommon and unbal-

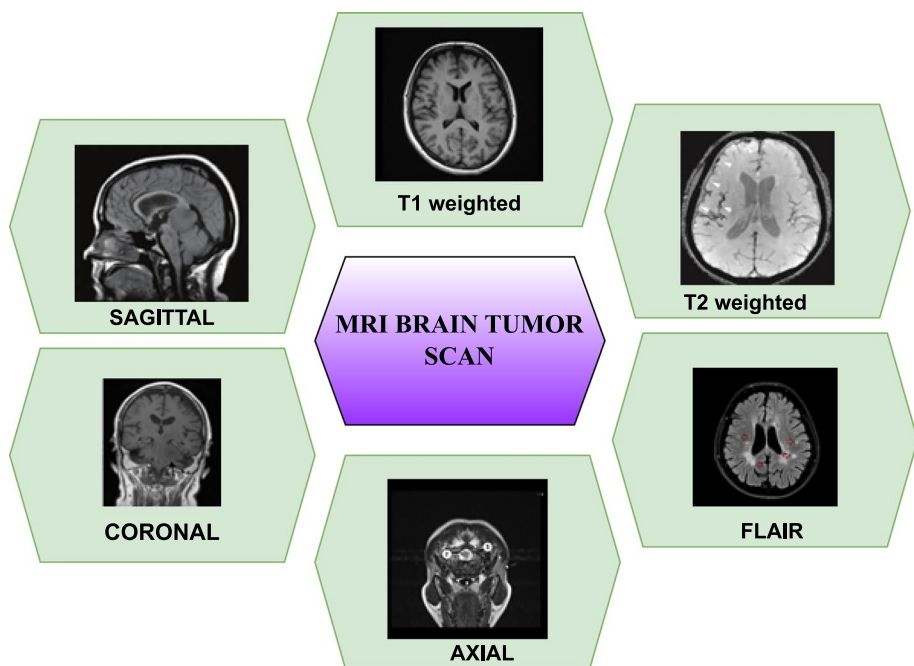


Fig. 6 Different MRI scan of brain tumor data

anced tumor types. Improving the generalization of uncommon tumor cases should require cooperative data-sharing initiatives with the advancement of few-shot learning.

Class imbalance in brain tumor detection significantly affects the accuracy and overall performance of AI models. Key impacts include:

- *Bias toward majority classes* Models tend to be biased toward the majority class (e.g., no-tumor cases) leading to poor sensitivity (recall) for minority tumor types such as Glioma or Meningioma. This can cause high false negative rates, missing actual tumor cases.
- *Reduced detection accuracy for minority classes* The minority tumor classes often have fewer samples, impairing the model's ability to learn distinctive features. This results in lower precision, recall, and F1 scores for these classes compared to well-represented classes like Pituitary tumors or no tumor cases.
- *Misclassification rates* Imbalanced datasets can increase misclassification rates specifically in rare tumor categories due to insufficient training examples, impacting clinical reliability of diagnostics.
- *Model overfitting to majority class* Without balancing techniques, models may overfit to majority class characteristics and fail to generalize well, reducing robustness across diverse patient data.

In conclusion, addressing class imbalance is crucial for achieving high and clinically reliable accuracy in brain tumor detection AI models by ensuring fair and effective learning across all tumor types and healthy cases

3.5.7 Ethical and legal considerations in MRI studies

The use of AI-based analytic models with MRI scans raises complex moral, ethical, and legal issues, primarily concerning patient privacy and potential algorithmic bias. It has to be ensured that the AI model is not only efficient, reasonable, and accurate but addresses important legal and ethical issues in MRI research about AI-based brain tumor detection. When applying AI systems in clinical decision-making, ethical challenges become especially significant in cases involving rare tumors. The consequences of this type of issue are noted as overdiagnosis, false positives, and missed diagnoses. The AI model does not train through rare cases and results are unable to identify rare tumors. This can result in delayed or failed early-stage treatment. AI diagnoses are further complicated by inadequate or unrepresentative data, which can negatively impact subsequent medical decisions, including operations and treatments. Therefore, ethical oversight, along with awareness of the model's limitations, is essential to address these issues effectively. Ethical AI in healthcare is designed to take responsibility for patient safety by minimizing risk and ensuring compliance with legal requirements.

Ethical concerns in AI-based diagnostics can be resolved through a combination of regulatory, technical, and operational strategies:

- *Patient consent and transparency* Patients must be clearly informed about how AI is used in their diagnosis and treatment, what data is collected, and any potential risks. Full transparency includes explaining how the AI system works and what role it plays

in medical decisions.

- *Bias and fairness* Rigorous evaluation and validation of AI algorithms should be performed on diverse, representative datasets to minimize bias. Independent third-party audits and ongoing monitoring can help ensure that algorithms are fair and equitable for all patient groups.
- *Accountability and liability* Clearly defined legal frameworks should determine responsibility in case of errors or harm caused by AI systems, potentially through joint liability between developers, healthcare providers, and institutions. Developers must certify their products' safety and efficacy, while providers must use them appropriately.
- *Data privacy and security* Strong privacy safeguards must be applied, including data minimization, anonymization, and privacy-by-design features in AI models. Informed consent for data use and clear rules about data ownership are essential.
- *Regulation and oversight* Implementation of strict regulatory standards, independent audits, reporting guidelines specific to AI diagnostics (like STARD-AI), and ongoing performance monitoring help ensure ethical compliance, safety, and effectiveness.
- *Continual improvement* Regular updates to algorithms, ongoing research, and adaptation of legal systems are necessary to address new ethical challenges as AI evolves.

Applying these principles creates a trustworthy environment for the use of AI in diagnostics, supporting patient rights, safety, and ethical medical practices.

4 Discussion

4.1 AI model for brain tumor detection and classification

Brain tumors can have deadly significance and need costly, risky actions, as Mahfuz et al. described. Patients in certain regions suffer from excruciating pain and adverse outcomes, which emphasizes the crucial need for better methods of discovery and cataloging. To obtain accurate info or data about brain tumor cases, primary data collection engages in direct interactions with the public and healthcare earners. Having a major impact is the primary goal (Sadeghi et al. 2024). The main of this research is to describe the various challenges related to managing to detect brain tumors and also improve the uncovering and classifying the accuracy-related through the use of hybrid deep learning methods. This all-encompassing strategy aims to enable early interventions, improve performance accuracy, and improve the quality of life for people with brain tumors in Faridpur and the surrounding areas. The structure of the system consists of several layers and movable parameters. Each MRI image is processed by GRU layers after a series of ViT layers. To develop a comprehensive tool for accurate brain tumors in medical imaging, we plan to use these components to improve temporal modeling detection and diagnosis feature representation for trustworthy termination decision-making.

CNN is a crucial tool that shares many physical and practical comparisons with biotic and graphic schemes, as demonstrated by its shared image classification task. Rustom et al. (2023) The most straightforward way to monitor a CNNs performance on a tumor discovery and classification job across several trials is to use an exactness metric (Sinha et al. 2023). The testing accuracy of a network is determined by dividing the number of correctly secret

images by the total number of instances of that category in the testing set that were kept out of the network during the exercise. The use of animal transfer learning in camouflage offers a practical way to compare performance before and after. However, testing accuracy does not account for the impact of training on the internal state of the network. However, testing accuracy does not explain how training impacts the internal state of the network. This leads to the well-known black box problem in artificial intelligence, where the model's operation is unclear, making it difficult to understand how a specific result was achieved (Khater et al. 2023). To counter this, we combine three specific XAI analysis methods with accuracy metrics in a unique way: feature spaces, DDI, and copy saliency mapping. These techniques allow us to examine the effects of brain tumor training and the proposed novel transfer learning regimen on the models (Shukla et al. 2024).

Another method for viewing the inner state of an accomplished CNN is to use a feature visualization technique called DDI. For each category of objects, this algorithm generates a visual prototype or feature print by recognizing and improving particular visual patterns that the system finds in the input data during training. In this prototype, the inner representation of each group by the network is essentially shown directly. However, image saliency maps, sometimes referred to as sensitivity maps, highlight the most salient features of each image. This technique provides insight into how a system makes cataloging decisions by overlaying an image with a heat map that highlights the areas of the input image that are most vital for sorting.

Hanaa zainEldin et.al identifying a brain tumor requires time, and the radiologist's knowledge is essential. Outdated practices are now costly and ineffective due to the growing amount of data that must be managed along with the patient population. Numerous researchers looked into various algorithms that were quick and accurate in identifying and categorizing brain tumors (Zangana et al. 2024). To develop automated systems that can rapidly and precisely identify or segment brain tumors, DL techniques have newly become more and more popular. In particular, DL makes it possible to classify brain cancers using medical images using a pre-trained CNN model. For the suggested CNN-based brain tumor classification model (BCM-CNN), the CNN hyperparameters optimization is carried out using the algorithm (ADSCFGWO). After enhancing hyperparameters, a training model called Inception-ResnetV2 is built. In general, hyperparameters fall into two groups: (i) those that define the network's basic structure, and (ii) those that control network training. The ADSCFGWO algorithm incorporates both the sine cosine and grey wolf algorithms, leveraging their respective strengths within a versatile framework (Nisar et al. 2021). The experiment's findings show that the BCM-CNN classifier outperformed the others due to the CNNs enhanced presentation due to the CNN optimization's hyperparameters. With the BraTS 2021 dataset, the BCM-CNN achieved 99.9% accuracy.

Saurabh Mandli, the most prevalent and debilitating issue in modern times is brain tumors. A tumor's late detection results in the death of numerous people every day; these deaths could have been prevented if the tumor had been discovered sooner. The complicated structure and uncontrollable growth of the tumor make early diagnosis extremely difficult. The development of CNNs and pre-trained models has led to the growth of numerous tumor detection replicas in recent decades. A significant lack of explanation and interpretability of the model is evident in the majority of the solutions, which only concentrate on accuracy (Jacqueline and Koppaiyan 2024).

Using pre-trained models, this work proposed an understandable perfect for brain tumor detection and classification. Four phases make up the proposed model. The first step involves creating synthetic MRI images of different classes using a provisional reproductive argumentative network (cGAN) to handle data overfitting and unbalancing. Brain tumors are identified in the second phase using various pre-trained representations, such as Mobile Net, Inception ResNet, Efficient Net, and VGGNet. If the tumor is identified, it is classified in the third phase using additional pre-trained models. In the final step, the model result is interpreted using LRP (GTIFA et al. 2024).

Mohamed Musthafa et al, the ability of MRI to produce detailed, high-contrast images has made it a focus of research on brain tumor detection using their picture representation. The normal and abnormal brain tissues have been distinguished using a range of traditional image processing techniques, including feature extraction and segmentation. Various types of techniques are frequently used followed by different constraints. Based on the recent developments in analysis the automatic techniques for detecting brain tumors are used by ML algorithms. It supports growth and diagnosis effectively. Based on various instances most of the researchers used Random Forest Classifier and SVM, These are all implemented in MRI to identify the tumors. Machine learning techniques help to save time, but it is not supported by other types of datasets.

Deep learning methods differ from traditional machine learning, it is enabled for learning the feature followed by the physical judgment removal. These types of brain detection are faced with multiple types of issues focused on various demands. This research study mainly focused in to assess the model followed by performance testing, dataset, provides possible solutions. These are all helpful for the healthcare applications. The study intends to fill important gaps in the current state of medical imaging analysis by advancing a more transparent, intelligible, and trustworthy AI-based tool for brain tumor detection.

Umme Aeman, describes an algorithm for deep learning that uses various pre-trained neural network models to detect and categorize brain tumors (Yang et al. 2023). Datasets that are frequently used in the medical imaging industry were used to assess the model. All things considered, the findings imply that pre-trained neural network models work well for classifying and categorizing brain tumors, and the suggested model may prove to be a helpful resource for specialists in neuroimaging. It is crucial to remember that extra study and testing are required to confirm the findings and raise the models' accuracy (Nazir et al. 2023). To detect the tumor images, we are introduced to deep CNN as a proposed layer. Image Net dataset is applied through a test for evaluation with the proposed model. Based on the result we have to compare the other classification this model achieved with 98.87% accuracy, 99.2% specificity, and 99.5% sensitivity.

The author describes about detection of brain tumors using CNN models, A deep learning method that provides efficiency among to modify the fuzzy C clustering. To classify deep learning models are used by optimizers as clustering. The datasets BraTS2015, 2017 and 2018 are evaluated through the proposed model. In their optimized CNN framework, Ramtekkar et al. (2023) utilized a compound filter for preprocessing, followed by GLCM-based feature extraction and hybrid metaheuristic optimizers (whale and gray wolf algorithms). This approach achieved an impressive detection accuracy of 98.9% on a brain tumor MRI dataset. The work proposed in Elhanashi et al. (2023) presents a deep learning-based framework for the classification and localization of multiple chest X-ray abnormalities, integrating

detection and recognition in a unified model. Their approach demonstrates high accuracy and robustness, contributing to improved computer-aided diagnosis in medical imaging.

The diagnosis and detection of brain tumors is greatly aided by AI, a helpful addition to the infamously challenging field of brain tumor surgery. Neuropathology has undergone a radical change thanks to subsets of AI with ML and DL. Techniques like feature extraction, selection, reduction, classification, and data preprocessing involve a lot of steps. As per the author, AI has boosted neuropathologists' self-assurance in detecting brain tumors. Recent advances in deep learning have produced a wide range of practical uses in domains as diverse as pattern cataloging, object discovery, voice appreciation, and decision-making. Healthcare researchers have used a range of ML and DL algorithms, including SVMs, k-nearest neighbor (k-NN), Naive Bayes, trained CNNs, VGGNets, GoogLeNet, and ResNets, DT to diagnose cancer. Research progress is further hindered by the lack of complete medicinal datasets since patient information cannot be shared due to privacy concerns.

Table 4 describes the datasets are used to detect brain tumors, in these different types of data sets are used. Explainable AI techniques used in BraTs 2019 and 2021 data set as Grad-CAM, guided integrated gradients. The result of this segmentation was achieved by 84.10%. Data set BraTs 2020 used by explainable technique as ODE (neural ordinary differential equations), achieved by the accuracy rate as ET (explanation type) as 99.649%, TC (transparency context) as 99.425%, and WT (what to explain) as 99.383%. Grad CAM techniques used for data set BraTs 2018, accuracy rate as ET as 0.79%, WT as 0.91%, and TC as 0.85%. The explainable artificial intelligence technique (Grad Cam) used in BraTs 2019 and FeTS2022 resulted in 0.745% of ET, 0.782% of TC, and 0.882% of WT (Kumar and Jyoti 2024).

Key challenges in generalizing AI models include:

- *Dataset bias* If the training data is not diverse or representative of real-world scenarios, models tend to learn superficial or context-specific patterns, causing poor performance when exposed to new or different datasets.
- *Overfitting* Complex models may become too tuned to the training data—capturing noise or irrelevant details—leading to high accuracy on training data but poor results on unseen data.
- *Underfitting* On the other hand, overly simple models fail to capture important patterns, resulting in suboptimal learning and generalization across datasets.
- *Model complexity versus simplicity trade-off* Striking the right balance between a sufficiently complex model (to capture key data features) and a simpler model (to avoid overfitting) is a persistent challenge.
- *Scalability and adaptability*: As data size and feature diversity grow, scaling models while maintaining generalizability becomes increasingly difficult.
- *Domain shift and dataset shift* AI models may not adapt well to data distributions that

Table 4 Dataset used for explainable AI

Dataset	XAI	Results
BraTs 2020	ODE	ET—99.649%, TC—99.425%, WT—99.383%
BraTs 2018	Grad-CAM	ET—0.79, WT—0.91, TC—0.85
BraTs 2019 and FeTS 2022	Grad-CAM	ET—0.745, TC—0.782, WT—0.882

differ from those seen during training—for example, due to geographic, demographic, or environmental changes.

- *Ethical and fairness concerns* Models must generalize fairly to all relevant populations. Ensuring that generalization does not come at the cost of introducing bias is essential in sensitive domains such as healthcare or finance.

Effectively addressing these challenges is crucial for deploying robust AI models that work reliably across varied, real-world situations.

4.2 Performance comparison of proposed model to other classification models

This comparison table discusses the problems, approaches, algorithms, techniques, limitations, and results of existing and proposed papers. This review collected by 50 researchers comes from the 2020 to 2024. Table 5 shows the problem, Existing and proposed approaches, and results of these given problems. A data set for AI-based Brain tumor detection and classification typically followed by the various choices of medical images related to MRI scans followed by the labeled with tumor identifications. The Data sets are used as the trained AI model of deep learning neural networks, it is used for automatically detect and classify brain tumors. Here explained some common data set is used for tumor detection and classification with AI. Classification and division of brain tumors are the typical uses for the BraTS Dataset. It provides multi-model MRI scans followed by FLAIR, T1, T2 sequence, and T1c. These kinds of data sets contain the manual segment followed by the segment annotations commonly used as classification tasks. KBT data set contains brain MRI pictures of patients with Brain tumors (Faris et al. 2024). It is specifically designed for classification that comes under the model identified as present or not. This data set contains 2500 images. The format of the dataset is JPG format and provides the label through specific tumor types. BTID is used to train the AI representations for tumor detection and classifications. NCI is used for AI applications in medical imaging. ISLES should be adopted by certain types of datasets and involving the segmentation and classifications. MICCAI (2017) mainly contains the segmentation of brain tumors provides multi-model imaging with tumor annotations and is used for both segmentation and classification among the various tasks. REMBRANDT is used for the advanced types of classification tasks (Gyamfi et al. 2022). Such datasets are used in AI models to identify and categorize brain tumors through data augmentation and image processing methods like MRI. CNN, U-Net for segmentation, and ResNet or VGG for classifications are all included in deep learning models K. Nishanth Rao, 2024.

Brain tumor diagnosis remains a significant challenge in the medical field due to variations in tumor size, location, and type. Accurate identification and analysis are crucial for early detection and effective treatment. Various computational approaches and machine learning techniques, including CNNs, Gradient Descent, and Decision Trees, have been widely utilized to enhance diagnostic accuracy. Many researchers have adopted deep learning methods in combination with CT and MRI to improve tumor classification and segmentation.

Recent advancements in image processing, AI, and hybrid models have further refined tumor detection, leveraging pixel-based feature extraction and automated classification techniques. Studies have demonstrated 97.11% accuracy in tumor detection, while optimized frameworks utilizing deep learning models have achieved 98.67% accuracy. These

Table 5 Performance comparison of proposed model to other classification models

References	Problem	Algorithms/techniques	Limitation(s)	Results
Sinha et al. (2023)	Dependability and credibility in healthcare environments	Convolutional neural network, Gradient Descent, Decision tree	Minimum threshold size images	Predictions using image representation: green (negative), red (positive), white (no effect)
Solanki et al. (2023)	Utilizing brain MRI to locate and identify tumor-infected regions	Computational intelligence and statistical image processing techniques, Voxel clustering algorithm, FCM algorithm	Small portion of the tumor may be mistaken for a normal area	DSC for segmentation 97.91%, identified abnormal pixels with 98.67% accuracy
Marmolejo-Saucedo and Köse (2022)	Brain tumor diagnosis setup	CNN model, Grad-CAM	Minimum size images	96.81% specificity, 95.58% sensitivity, and 97.11% accuracy
Zeineldin et al. (2022)	Brain image analysis	Explainable AI of DLN using Neuro XAI framework	Methods of classification explanation limited to 2D	Radiologists and medical specialists identify and diagnose brain tumors in clinical procedures
Gaur et al. (2022)	Diagnostic issues	CNN-based DL, XAI algorithms (LIME, SHAP)	Limited dataset	Strong accuracy in image classification and recognition
Maqsood et al. (2022)	Clinical diagnosis	Modified MobileNetV2 architecture, 17 layers of deep neural networks, M-SVM, BraTS 2018	Technique used for 2-D MRI Images	Visual evaluation 97.47% accuracy, Quantitative evaluation 98.92% accuracy
Nazir et al. (2023)	Problems with the diagnosis of breast cancer, genitourinary, and cardiovascular	ML Algorithms	Classification and segment process	Prompt interventions improve patient health and safety
A. and R. (2023)	Real-world problems	XAI or White box	Unsuitable for models of global explanation	More precise without human assistance
Mehrotra et al. (2020)	Identifying complex medical disorders, such as a brain tumor	Deep Learning Algorithms	Long training time (112 min) using ADAM with GoogLeNet	Outstanding concert with maximum accuracy of 99.04%
Jena et al. (2022)	Computer vision problem	PRISMA search approach, Radio genomics approach	Minimum dataset	Influence on beneficial results of BTC's radio genomics strategy
Giuste et al. (2022)	Diagnosis and disease precision staging	PRISMA technology, XAI, Image segmentation analysis algorithms	Limited label dataset	XAI techniques improve patient care quality
Saeedi et al. (2023)	Labeling images	Two deep learning methods, ML approaches, 2D CNN	Small size of medical image database	Two-dimensional CNN classifies brain tumors with highest accuracy
Raza et al. (2022)	ReLU problem	DeepTumorNet	Less time complexity	F1 score of 99.66%, recall 100%, accuracy 99.67%, precision 99.6%
Rasool et al. (2022)	Medical problems	MRI, CNN-based architecture, CNN algorithm, Google Net model	Limited size	Finely tuned Google-Net model achieved 93.1% accuracy

Table 5 (continued)

References	Problem	Algorithms/techniques	Limitation(s)	Results
Gómez-Guzmán et al. (2023)	Multi-class problem	MRI	Kaggle stores research data	97.12% average accuracy
Rehman et al. (2020)	Medical problems	3D CNN	Limited dataset	Accuracy of 98.32, 96.97, 92.67
Kumar et al. (2022)	Medical problems	AI-based learning methods, CADx, CADe	Limited dataset	Significant results
Lima et al. (2022)	Biomedical-developing problem	CAD methodologies (PET, MEG, MRI)	Recover precise analysis	Precise analysis
Anaya-Isaza and Mera-Jiménez (2022)	Depth analysis	MRI, ResNet50, PCA-Based Augmentation	256x256 uint8 images	F1 score of 92.34%, significance level 0.05
Bhandari et al. (2022)	Health issue	SHAP, LIME, Grad-CAM, XAI framework	Numerous parameters increase resource needs	Average test accuracy $94.31 \pm 1.01\%$, validation accuracy $94.54 \pm 1.33\%$
Tummala et al. (2022)	Diagnosis of brain tumors	CNN models, MRI	spatial resolutions of either 512×512 or 256×256	VIT models test accuracy 98.7%, individual model L/32 98.2% test accuracy at 384×384
Aamir et al. (2022)	Manually diagnosis	MRI, CAD	Limited list	Overall accuracy 98.95%
Nawaz et al. (2022)	Inefficient diagnosis process	Hybrid-brain-tumor-classification (HBTC) framework	Less observational variability	HBTC framework performance 98.8%
Singh et al. (2020)	Medical imaging problems	CAD using AI	Limited medical reports	Interpretability and visual appeal
Ullah et al. (2022)	Global issue	TL, CNN	Limited dataset	Recall 99.75%, accuracy 98.91%, precision 98.28%
Kang et al. (2021)	Clinical diagnosis	MRI, SVM, RBF	Classification and segmentation	Selected features, 92% accuracy
Khan et al. (2021)	Psychological problems	KNN, SVM, PSO-based optimized SVM (SDP-SVM) approach	Distinguish AD from MCI patients using 18F-FDG-PET data	Very high detection accuracy
Zahoor et al. (2022)	Diagnosis and effective treatment	Kaggle, Figshare benchmark datasets, CNNs, MRIs	Manual feature extraction	DBFS-EC detection scheme achieves accurateness 99.56%, exactness 0.9991, remembrance 98%, F1-Score 99%, MCC 98%, AUC-PR 99%
Tjoa and Guan (2020)	Detect erroneous reasoning	LIME, Grad-CAM, Alex Net CNN, VGG	Clinical evidence dataset	Human subjects attained 61.23% accuracy, 16.79% better than guided backpropagation visualization
Ghaffar Nia et al. (2023)	Computer vision problem	SVM, CNN, RNN, LSTM networks, XmasNet	Manual data collection	97.057% accuracy

Table 5 (continued)

References	Problem	Algorithms/techniques	Limitation(s)	Results
Murthy et al. (2022)	Classification issues	Adaptive coefficient vector-based DHOA (ACV-DHOA), adaptive fuzzy deformable fusion (AFDF), FCM, DHOA	Clinical evidence data	High classification accuracy compared to conventional methods
Forghani (2020)	Clinical diagnosis problem	CNN, prediction algorithm construction	Clinic useful information	Significant progress
Tufail et al. (2021)	Multi-class classification using Long Short-Term Memory (LSTM)	Computer-aided design (CAD), Convolutional Neural Networks (CNNs)	Preprocessing-free basic processing of natural data	Fewer data manipulations and improved outcomes
Montaha et al. (2022)	Multi-class classification problem	Long Short-Term Memory (LSTM), Time Distributed-CNN-LSTM (TD-CNN-LSTM), 3D Convolutional Neural Network (CNN)	Employing 2D slices rather than 3D sequences, using only one sequence, and not experimenting with the various configurations of the suggested model	Achieving the highest test accuracy of 98.90%
Hakkoum et al. (2022)	Trade-off between accuracy and interpretability of a model	SVM, artificial neural networks (ANNs), decision trees (DTs), or linear classifiers	Interpretability and transparency	Experiment-based empirical type
Shah et al. (2024)	Mental illnesses	SVMs and CNN	Test subjects	Improve the predictability and reliability of such disease detection and prediction
Wagan et al. (2022)	Diagnose problems	IoMT, ANN, SVM, SHAP	Limited data size	By enhancing the everyday healthcare management system and utilizing IoMT devices, risk can be decreased
Rammurthy and Mahesh (2022)	Segmentation Problem-multi-class classification	Computer-aided diagnosis (CAD), optimization-driven technique, WHHO, Whale optimization algorithm (WOA), HHO	Followed by a data set	0.816, 0.791, and 0.974 as the maximum accuracy, specificity, and sensitivity, respectively
Khan et al. (2020)	Rarely addressed problems	TML and DL techniques in MIA, CAD system	Medical expertise and clinical knowledge	High-performance MIA models, results are accurate
Sharif et al. (2021)	Multiclass classification problem	Pre-Trained Deep Learning Model, EK-bHFV, modified genetic algorithm	Impact on the system's accuracy	The accuracy of BRATS2018 and BRATS2019 is over 95%
Kumar (2023)	Different image segmentation techniques	DWT, CNN, BRATS, optimal machine-learning models	Data set	Minimum and maximum contour values

Table 5 (continued)

References	Problem	Algorithms/techniques	Limitation(s)	Results
Diaz-Pernas et al. (2021)	Segmentation problem	Multiscale Convolutional Neural Network, T1-CE MRI image dataset, multi-pathway CNN architecture	Visual field	Average sensitivity of 0.940, average dice index of 0.828, and average ptas value of 0.967
Agrawal et al. (2022)	Dimensionality, generalization	3D U-Net model, CNN, classical neural network	Large training samples, 2D ANN, 3D volumetric data	Higher efficiency
Jun and Li-yuan (2022)	Large number of parameters	Multipath network, Network Optimization Algorithm, CNN, CAD	Focal point of the lesion	With an accuracy of 98.61%
Ahuja et al. (2022)	Multi-class classification	T1W-CE (MRI), DarkNet-19 and DarkNet-53	Medical imaging modalities	Best-performing pre-trained DarkNet model had a training accuracy of 99.60% and a validation accuracy of 98.81%
Albahri et al. (2023)	Security and privacy problems	bibliometric approach	Limited dataset	High transparency and dependability
Hassija et al. (2024)	Real-world problems	XAI, ImageNet image-classification, multidisciplinary approach	Compromise between precision and interpretability	Improved accuracy of predictions and increased transparency

high-precision models rely on enhanced feature extraction, noise reduction techniques, and advanced neural networks, making them highly effective in medical imaging analysis.

Despite these advancements, challenges such as computational complexity, false positive rates, and model generalization remain critical areas for further research. Future improvements may focus on XAI, transfer learning, and multi-modal data fusion to enhance interpretability and reliability in real-world clinical applications.

4.3 Evaluation metrics

The model's performance is assessed using accuracy, precision, recall, and F1 score which are essential for determining the effectiveness of deep learning models in tumor detection.

$$Accuracy = \frac{TP + TN}{TN + TP + FP + FN} \quad (1)$$

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

$$F1 - Score = 2 \times \frac{Precision \times recall}{Precision + recall} \quad (4)$$

where,

True Positive (TP) the prediction outcome is true, and its true in reality, also.

True Negative (TN) the prediction outcome is false, and is false in reality, also.

False Positive (FP) the prediction outcomes are true, but they are false in actuality.

False Negative (FN) the prediction outcomes are false, but they are true in actuality.

Figure 7 describes the various algorithms used to detect brain tumors successfully. ResNet 50 detects the accuracy as 96.78, precision as 52, Recall as 52, and F1 score as 48. CNN's accuracy rate is 96.03, precision is 98, recall is 97, and F1 score is 98%. VGG followed by accuracy as 97.09, Precision as 95, Recall as 92, and F1 score as 93. The inception algorithm was followed by an accuracy of 97.09, a Precision of 95, a Recall of 92, and an F1 score of 93.

Table 6 describes about performance analysis of the transfer learning model followed by ResNet 18, ResNet 50, VGG 16, VGG 19, efficient Net V21, Mobile Net V2, Dense Net 121, Alex Net, and ViT GRU. To compare all result scores followed by the precision, recall, and F1 score. After analysis, we suggest that most of the score values are similar to the optimizer. Each dataset has a training loss and validation loss. Training time is noted as Sec and Training and Testing accuracy as percentage format (Ahmed et al. 2024).

Figure 8 indicates that the ResNet family typically offers the optimal balance between speed and accuracy. ResNet-18 obtains very low loss values (0.011–0.012) with quick training times of roughly 6 s, however it occasionally experiences instability (loss spikes to 1.412). ResNet-50 is slightly slower (around 13 s), but it consistently gives the modest loss values (0.043–0.052), with only one outlier (0.534). The VGG models (16 and 19) are slower, taking 18–21 s, and produce greater and more inconsistent loss values (0.052–0.559), making them less dependable. EfficientNet-V2 strikes a balance between modest

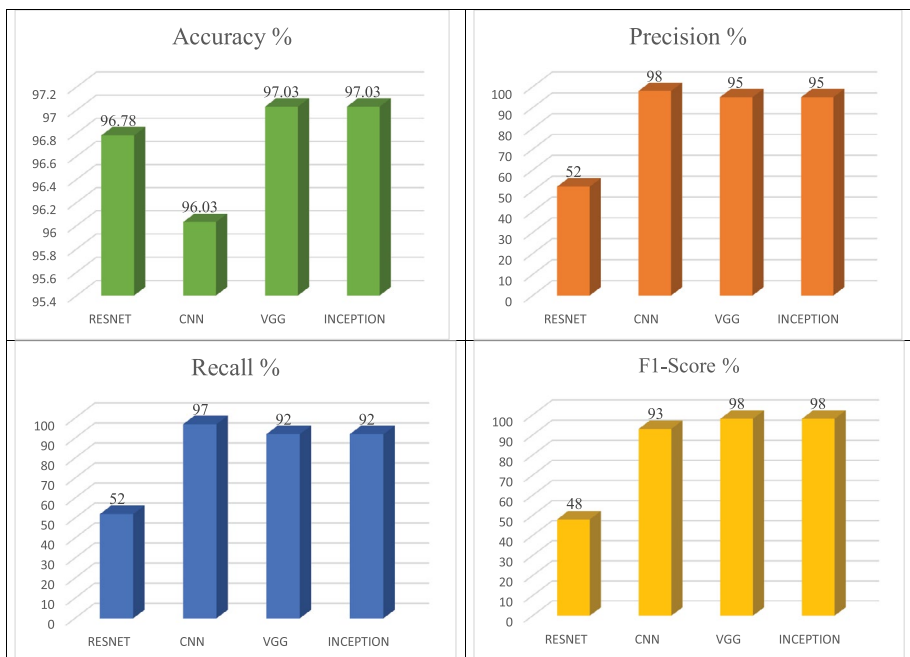


Fig. 7 Metrics among different models

Table 6 Performance analysis of the transfer learning model detection

Model	Optimizer	P* (%)	R* (%)	F1* (%)	TL* (%)	VL* (%)	TT* (s)	TA* (%)	VA* (%)
ResNet 18	Adam W	91	91	90	0.011	0.157	5.83	99.88	94.29
ResNet 18	Adam	94	93	93	0.012	0.119	6.07	99.88	96.86
ResNet 18	SGD	65	81	72	1.412	1.479	5.63	81.25	80.57
ResNet 50	Adam W	91	91	91	0.043	0.221	12.94	98.65	91.43
ResNet 50	Adam	92	92	92	0.052	0.245	13.03	98.65	92.86
ResNet 50	SGD	63	79	79	0.534	0.544	12.64	81.74	79.43
VGG 16	Adam W	91	92	91	0.052	0.152	18.41	97.43	93.71
VGG 16	Adam	93	93	93	0.534	0.141	18.9	99.02	94.00
VGG 16	SGD	72	81	75	0.054	0.534	17.73	78.55	81.43
VGG 19	Adam W	92	91	91	0.559	0.175	20.92	98.28	94.29
VGG 19	Adam	91	91	91	0.054	0.175	20.92	98.90	94.29
VGG 19	SGD	72	79	73	0.054	0.560	20.55	78.80	78.29
Efficient Net V2 I	Adam W	91	91	91	0.595	0.164	17.36	96.56	92.65
Efficient Net V2 I	Adam	89	89	89	0.072	0.295	14.61	95.83	91.71
Efficient Net V2 I	SGD	68	76	72	0.113	6.026	14.57	50.86	51.14
Mobile Net-V2	Adam W	85	86	85	6.024	0.336	6.43	90.2	85.14
Mobile Net-V2	Adam	80	83	80	0.288	0.364	6.54	88.97	83.71
Mobile Net-V2	SGD	74	84	77	0.281	2.505	11.82	82.27	84.86
Dense Net 121	Adam W	97	97	97	3.624	0.089	13.87	99.88	97.71
Dense Net 121	Adam	93	93	93	0.018	0.154	13.93	99.88	96.29
Dense Net 121	SGD	72	83	73	0.031	2.605	13.82	80.27	82.86
AlexNet	Adam W	86	87	86	2.724	0.239	4.57	92.89	89.43
AlexNet	Adam	89	89	88	0.198	6.849	4.05	96.08	91.14
AlexNet	SGD	71	84	77	0.145	0.005	4.29	79.78	84.00
ViT-GRU	Adam W	97	97	97	6.854	0.008	47.93	99.93	98.97
ViT-GRU	Adam	96	95	95	0.008	0.102	49.28	99.75	96.56
ViT-GRU	SGD	66	81	73	0.469	0.460	48.3	81.02	81.66

P*=Precision, R*= Recall, F1*= F1-Score, TL*=Training loss, VL*=Validation loss, TT*=Training Time, TA*=Training Accuracy, VA*=Validation Accuracy

training times (14–17 s) and minimal losses (0.072–0.113), while it occasionally exhibits instability (0.595). MobileNet-V2 is one of the fastest (6–12 s), however its performance is very inconsistent, with scores ranging from very poor (6.024) to fairly acceptable (0.281–0.288). DenseNet-121 shows great potential, achieving very low loss values (0.018–0.031) at moderate speeds (around 14 s), but it is not always consistent, with spikes as high as 3.624. AlexNet is the fastest model overall (4–5 s) with a better performance (loss about 0.145–0.198), however it occasionally performs worse (2.724), indicating that it is based on an older architecture. Lastly, ViT-GRU's computational expense (48–49 s) and significant instability (results ranging up to 6.854) limit its applicability, despite the fact that it achieves a good best-case loss (0.008). Overall, ResNet-18 and ResNet-50 are the best options because they have minimal loss and reasonable calculation time. EfficientNet-V2 and DenseNet-121 both look good, but they require some tweaking to keep them stable. MobileNet-V2 and ViT-GRU can be fast or highly accurate in certain cases, but their performance lacks consistency. AlexNet offers speed with moderate accuracy, while VGG models are the least preferred due to their slower performance and fluctuating loss values.

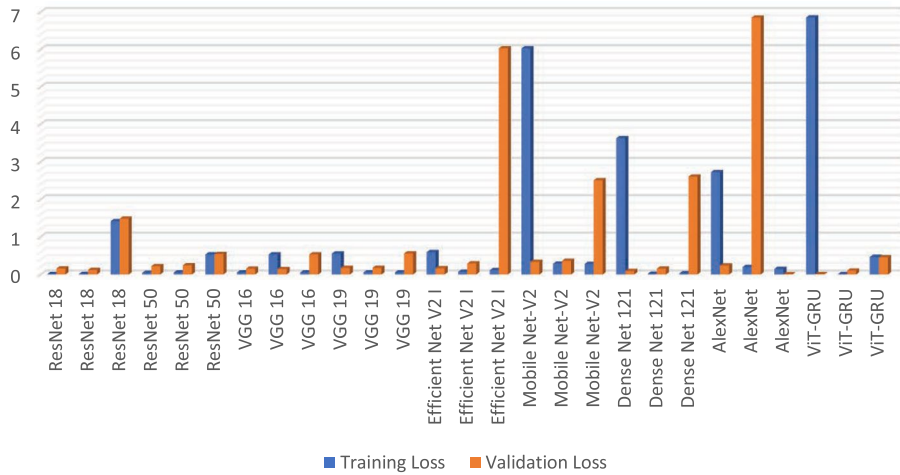


Fig. 8 Training and validation loss among different models

5 Results

The importance of XAI in medical imaging has increased., particularly for requests such as brain tumor detection. For medical specialists to use AI models in clinical settings, they must be able to believe and comprehend the AI's decision-making process, which makes the interpretability of AI models essential. The main conclusions and debates surrounding explainable AI for brain tumor detection are listed below.

5.1 Improved trust and adoption

Clinical confidence is the ability of clinicians to have faith in the predictions made by AI models is one of the primary obstacles to its application in healthcare. Through transparency, explainable AI enables physicians to comprehend how and why a model came to a specific conclusion. Better adoption of AI systems can result from this substantial increase in clinical confidence (Sadeghi et al. 2024). Ethical and legal aspects, it is AI-based medical decisions must be comprehensible, especially when they can potentially change a patient's life. By offering a transparent justification for tumor identification, classification, or treatment choices, XAI ensures accountability while addressing ethical and legal issues.

5.2 Visualizing AI's focus areas

Saliency Maps contains various Methods such as saliency maps (e.g., Grad-CAM) are frequently employed in the background of brain tumor detection to indicate which regions of an MRI or CT scan the AI model concentrates on during prediction (Marmolejo-Saucedo and Köse 2022). For example, radiologists can correlate the AI's decision with human interpretation by using saliency charts to visualize areas of the brain scan that correspond to normal tumor growth. Heatmaps are used for to create to display the areas of the brain scan that have the highest bearing on the finding of a tumor. When radiologists use a heatmap to

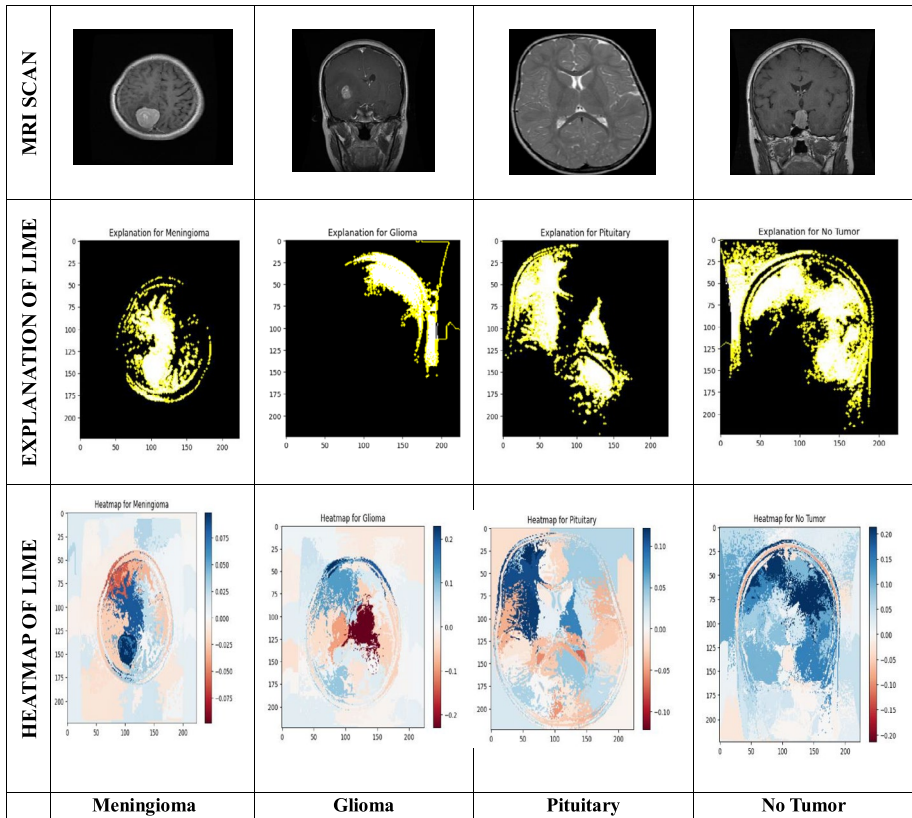


Fig. 9 MRI and lime XAI of brain tumor

identify tumor locations, it facilitates diagnosis and indicates whether the AI’s results are consistent or not. Locally approximating the model, the LIME method provides insights into which features were most important for a given decision, thereby aiding in the explanation of AI predictions. LIME can identify the areas of an MRI scan that the AI considers significant for its prediction to detect brain tumors as shown in Fig. 9.

5.3 Feature attribution

Finding relevant features explainable AI methods can determine which aspects of the MRI image, such as texture, shape, contrast, or intensity, were most important in identifying the tumor. These insights can help guide feature selection during training or propose new features that could increase model accuracy, which can enhance model performance (Lima et al. 2022). To Understanding Tumor Types, XAI can assist in identifying which features of the tumor’s appearance in MRI scans led the AI to categorize the tumor in a specific category for tasks involving tumor classification such as glioma vs. meningioma. A meningioma may be recognized by its smoother borders, whereas a glioma may be categorized by its irregular shape (Aamir et al. 2022).

5.4 Model explainability with deep learning

Challenges with Black-box Models: It is CNN, a type of deep knowledge model, is known to notice brain tumors with high accuracy. However, they frequently function as black-box models, meaning that their predictions are not completely explained. To interpret CNN predictions, XAI techniques such as Grad-CAM and SHAP assign output decisions to specific features of the input copy. To assign significance scores to the dissimilar pixels of the input image, deep neural networks employ the well-liked XAI technique known as LRP. This increases radiologists' transparency by elucidating the specific MRI scan components that most influenced the prediction (Hu et al. 2024).

5.5 Enhanced performance and error diagnosis

Error analysis and model debugging, by using XAI tools, one can spot potential errors in the AI model's tumor detection or classification. Developers can enhance the model by improving training, augmenting data, or modifying the model architecture by knowing which regions the model missed or misclassified (Jena et al. 2022). Bias and Fairness in Inadequate training of AI models may result in a bias toward specific tumor types or patient demographics. To produce more equitable and accurate predictions, XAI can detect and reduce these biases by indicating whether and where the model's predictions are based on skewed or irrelevant data (Albahri et al. 2023).

5.6 Patient-centric explanations

Personalized insights, which means each patient can receive a customized explanation of the tumor detection and classification results using XAI techniques (Bukhari and Mohyud Din 2022). When presenting results to patients, this is crucial because physicians can use these explanations and visualizations to give patients concise, intelligible information about their diagnosis. Empowering Patients by presenting patients with clear visuals such as annotated MRIs or tumor location heatmaps, AI predictions can help them better understand their medical condition. This method facilitates collaborative decision-making and gives patients the confidence to inquire more intelligently about their available treatment alternatives (Ramakrishnananda and Sarmah 2024).

6 Challenges and limitations

Balancing interpretability and accuracy, which means interpretability and model accuracy are frequently traded off. Although they are less interpretable, extremely complex models, such as deep learning networks, can achieve high accuracy. Aiming for more explainable AI models, such as hybrid approaches or simpler replicas like decision trees or rule-based models, may increase transparency while decreasing accuracy (Shoaip et al. 2021). Tumor characteristic complexity, a brain tumor's location, size, and shape can vary widely. Because of this intricacy, explainable AI methods may find it challenging to offer precise, reliable explanations. Research on creating reliable methods for this degree of complexity is still ongoing. Overhead in computation. Certain XAI techniques, like SHAP and LIME, can be

calculated as costly, particularly when examining sizable medical datasets. This may restrict their ability to be used practically in real-time clinical settings where efficiency is supreme (Borys et al. 2023).

The computational overhead of XAI methods can be reduced through several strategies:

- *Dimensionality reduction* Applying data preprocessing techniques like principal component analysis (PCA) or feature selection to reduce the number of input features before explanation reduces the computational burden without significantly affecting interpretability.
- *Sampling optimization* Reducing the number of samples used in approximation techniques such as SHAP and LIME through intelligent methods like clustering-based or adaptive sampling balances accuracy and efficiency.
- *Efficient surrogate models* Using lightweight surrogate models (e.g., sparse linear models or simplified decision trees) in place of complex approximations can lower computational costs in post-hoc explanation methods like LIME.
- *Hybrid methods* Combining inherently interpretable models with XAI applied only when necessary helps limit overhead. For example, simple interpretable models can be used routinely, while complex methods like SHAP or LIME are reserved for high-risk cases requiring detailed explanations.
- *Parallelization and hardware optimization* Utilizing modern hardware such as GPUs or TPUs and parallel computing frameworks accelerates computation of feature attributions in real-time applications.
- *Lightweight and real-time models* Designing and using XAI techniques specifically optimized to run on resource-constrained devices or in real-time, such as backpropagation-based methods that require a single pass, can minimize overhead.

Collectively, these approaches help make XAI more practical for deployment in real-time and resource-limited environments by reducing latency and resource consumption while maintaining explanation quality.

6.1 AI models with built-in interpretability

Attention Mechanisms, certain AI models are built with attention devices that enable them to essence on particular areas of an image (He et al. 2023). These mechanisms, which are frequently incorporated into convolutional layers, can be regarded as a type of explainability since they enable the model to identify important regions such as tumor regions during classification. Rule-based models used for the supplement deep learning, XAI can be used with more straightforward rule-based or decision-tree models to produce outputs that are easy to understand. Although these models may not be as precise as deep learning models, they provide clear decision-making procedures (Mehta et al. 2022).

6.2 Use of explainable AI in clinical trials and research

Regulatory approval AI models used in medicine, especially those that detect brain tumors, frequently have to adhere to stringent regulatory standards. These requirements can be met with the use of explainable AI, which supports assertions of model safety and depend-

ability by demonstrating the decision-making process (Yurtsever et al. 2024). Cooperative research, which means AI researchers, radiologists, and oncologists can work together more effectively thanks to XAI's transparency. By critically examining AI-generated results, clinicians can offer input and improve the model's practicality.

Various XAI algorithms applied in brain tumor detection exhibit unique advantages and trade-offs. Saliency-based methods such as Grad-CAM and CAM generate visual heatmaps that highlight tumor regions, making them intuitive for clinicians, but they are often sensitive to image noise and lack fine-grained fidelity. Perturbation-based approaches like LIME and SHAP provide more model-agnostic explanations by quantifying feature contributions, yet they are computationally expensive and less suited for real-time clinical use. Prototype-based and rule-based models improve interpretability through human-readable logic but may compromise predictive accuracy. Overall, there is a trade-off between interpretability, computational efficiency, and faithfulness to the underlying model. The choice of XAI method must therefore be guided by the clinical application context, whether prioritizing real-time deployment, detailed feature attribution, or transparent decision pathways for regulatory approval.

Explainable AI for brain tumor detection may completely change how radiologists and physicians work with AI systems. It enhances understanding, trust, and transparency, all of which are critical in a delicate area like medical diagnostics. XAI techniques will be used to help develop more dependable, accountable, and patient-centered AI systems as AI continues to play a bigger role in healthcare. Here it manages the various types of issues as complex and contains the tumor characteristics and efficiency. Hence, exploitation of XAI to detect brain tumors and to conduct more research followed by the various types of implementation and healthcare environments shows an increasing trend nowadays.

7 Conclusion and future scope

Artificial intelligence and pixel representation are used for explainable AI to detect brain tumors. Both models are important to check the accountability for trust in the healthcare industry. AI application is supported to take results followed by high performance and transparency. It is easy to understand the model classifier tumor in a specific way. Here various types of datasets such as enhanced diagnostic precision, the massive dataset of medical images. After completing the analysis, we got the result as a faster and more accurate one. Traditional types of AI should be represented as Block boxes, it is easy to make the specific results. XAI results are transparent, and most of the clinical advisers followed by AI judgments.

The findings indicate that ResNet 18 performed extremely well, achieving 94% training accuracy, 96.86% testing accuracy, a low loss of 0.012, and a quick runtime of about 6 s. ResNet 50 was marginally slower at around 13 s but remained stable, with a test accuracy of 92.86%. DenseNet 121 (Adam W) attained the highest accuracy at 97.71%, although its consistency varied across different optimizers. ViT-GRU also reached approximately 97% accuracy, with a very low loss of 0.008, it took approximately 49 s to complete the computation. Conversely, VGG models (roughly 94% test accuracy) and MobileNet-V2 (loss up to 6.024) proved less reliable, despite their faster training times. The computational cost of transformer-based models such as ViT-GRU should be reduced in future study. DenseNet

and EfficientNet need to be stabilized across different optimizers. To increase their generalizability, these models should be tested on large-scale, multi-institutional datasets. Furthermore, developing XAI approaches such as saliency maps, LIME, and SHAP will be critical for improving interpretability, boosting clinician trust, and ensuring ethical and patient-centered AI adoption in healthcare.

The recent growing use of AI in medical decision-making included various types of ethical and legal issues including liability for all bias. XAI model helps to address the various types of issues. AI systems should produce results independently, without requiring human supervision. This approach enhances clarity in explanations. Recent advancements aim to improve model accuracy, making it more reliable for diagnosing all types of brain tumors while incorporating patient information. Finally, we suggest that XAI should be used to detect and classify brain tumors, and also provide treatment for particular ones. It easily helps to provide a positive relationship to the clinical applications. According to that in a successful healthcare center providing the AI model, there is a balanced relationship between the compensation of automated decision-making and patient comprehension along with supervision.

Future research should focus on developing large-scale, multi-center datasets to ensure generalizability, as most current studies rely on limited or single-center data. Integrating multimodal information such as MRI, genomic data, and patient history could enhance diagnostic precision. Lightweight and real-time XAI approaches are also needed for clinical deployment. Ethical and legal challenges—such as bias, accountability, and regulation—remain unresolved and must be addressed. Furthermore, prospective clinical trials are crucial to assess real-world patient outcomes and to enhance clinician trust in AI-assisted tumor detection.

Author contributions L. A.: conceptualized the study, methodology, collected the data and wrote the main paper G. P.: performed review, editing and supervision G. V.: performed review, editing and supervision N. J.: Performed Review and editing

Funding Open access funding provided by Vellore Institute of Technology. This research received no specific grant from any funding agency, commercial, or not-for-profit sector.

Data availability No datasets were generated or analysed during the current study.

Declarations

Ethical approval This study did not involve human participants or animals, and therefore ethical approval was not required.

Conflicts of interest The authors declare no Conflict of interest.

Consent for publication All authors have given their consent for publication.

Materials availability Not applicable.

Code availability Not applicable.

Conflict of interest The authors declare no Conflict of interest.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License,

which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Aamir M, Rahman Z, Dayo ZA, Abro WA, Uddin MI, Khan I, Hu Z (2022) A deep learning approach for brain tumor classification using MRI images. *Comput Electr Eng* 101:108105. <https://doi.org/10.1016/j.compeleceng.2022.108105>
- Abdusalomov A, Mukhiddinov M, Whangbo TK (2023) Brain tumor detection based on deep learning approaches and magnetic resonance imaging. *Cancers* 15:4172. <https://doi.org/10.3390/cancers15164172>
- Aeman U, Kaleem M, Sarwar M, Mushtaq M, Choudhary S, Rashid S, Azhar M (2023) A systematic literature review on classification of brain tumor detection. *J Comput Biomed Inf* 5:327–337
- Agrawal P, Katal N, Hooda N (2022) Segmentation and classification of brain tumor using 3d-UNet deep neural networks. *Int J Cogn Comput Eng* 3:199–210. <https://doi.org/10.1016/j.ijcce.2022.05.003>
- Ahmed M, Hossain M, Islam MR, Ali M, Ahmed M, Ahmed K, Miah MS, Rahman M, Niu M, Islam M (2024) Brain tumor detection and classification in MRI using hybrid ViT and GRU model with explainable AI in southern Bangladesh. *Sci Rep* 14(1):22797. <https://doi.org/10.1038/s41598-024-71893-3>
- Ahmed MM, Hossain MM, Islam MR, Ali MS, Nafi AAN, Ahmed MF, Islam MK (2024) Brain tumor detection and classification in MRI using hybrid ViT and GRU model with explainable AI in southern Bangladesh. *Sci Rep* 14(1):22797. <https://doi.org/10.1038/s41598-024-48702-w>
- Ahuja S, Panigrahi BK, Gandhi TK (2022) Enhanced performance of dark-nets for brain tumor classification and segmentation using colormap-based superpixel techniques. *Mach Learn Appl* 7:100212. <https://doi.org/10.1016/j.mlwa.2022.100212>
- Albahri AS, Duhaime A, Fadel M, Alnoor A, Baqer N, Alzubaidi L, Albahri OS, Alamoodi A, Bai J, Salhi A, Santamaría J, Ouyang C, Gupta A, Gu Y, Deveci M (2023) A systematic review of trustworthy and explainable artificial intelligence in healthcare: assessment of quality, bias risk, and data fusion. *Inf Fusion* 96:156–191. <https://doi.org/10.1016/j.inffus.2023.03.008>
- Ali M, Gilani S, Waris M, Zafar K, Jamil M (2020) Brain tumour image segmentation using deep networks. *IEEE Access*. <https://doi.org/10.1109/ACCESS.2020.3018160>
- Anaya-Isaza A, Mera-Jiménez L (2022) Data augmentation and transfer learning for brain tumor detection in magnetic resonance imaging. *IEEE Access* 10:23217–23233. <https://doi.org/10.1109/ACCESS.2022.3153999>
- Balwant M (2022) A review on convolutional neural networks for brain tumor segmentation: methods, datasets, libraries, and future directions. *IRBM* 43(6):521–537. <https://doi.org/10.1016/j.irbm.2022.05.002>
- Bhandari M, Shahi TB, Siku B, Neupane A (2022) Explanatory classification of CXR images into COVID-19, pneumonia, and tuberculosis using deep learning and XAI. *Comput Biol Med* 150:106156. <https://doi.org/10.1016/j.compbimed.2022.106156>
- Borys K, Schmitt Y, Nauta M, Seifert C, Krämer N, Friedrich C, Nensa F (2023) Explainable AI in medical imaging: an overview for clinical practitioners—beyond saliency-based XAI approaches. *Eur J Radiol* 162:110786. <https://doi.org/10.1016/j.ejrad.2023.110786>
- Bukhari ST, Mohy-ud-Din H (2022) EID3 U-net for brain tumor segmentation: submission to the RSNA-ASNR-MICCAI BraTS 2021 challenge, pp 276–288. https://doi.org/10.1007/978-3-031-09002-8_25
- Chen M, Linton-Reid K, Aboagye EO, Copley SJ (2025) Translating radiomics into clinical practice: a step-by-step guide for study design and evaluation. *Clin Radiol*. <https://doi.org/10.1016/j.crad.2025.107053>
- Diaz-Pernas FJ, Martínez-Zarzuela M, Antón-Rodríguez M, González-Ortega D (2021) A deep learning approach for brain tumor classification and segmentation using a multiscale convolutional neural network. *Healthcare* 9(2):153. <https://doi.org/10.3390/healthcare9020153>
- Elhanashi A, Saponara S, Zheng Q (2023) Classification and localization of multi-type abnormalities on chest x-rays images. *IEEE Access* 11:83264–83277. <https://doi.org/10.1109/ACCESS.2023.3302180>

- Farhan AS, Khalid M, Manzoor U (2025) XAI-MRI: an ensemble dual-modality approach for 3D brain tumor segmentation using magnetic resonance imaging. *Front Artif Intell* 8:1525240. <https://doi.org/10.3389/frai.2025.1525240>
- Faris RA, Mosa Q, Albaidri M (2024) Robust classification for sub brain tumors by using an ant colony algorithm with a neural network. *J Wirel Mobile Netw Ubiquitous Comput Dependable Appl* 15(2):270–285. <https://doi.org/10.58346/JOWUA.2024.12.018>
- Forghani R (2020) Precision digital oncology: the emerging role of radionics-based biomarkers and artificial intelligence for advanced imaging and characterization of brain tumors. *Radiol Imaging Cancer* 2(4):190047. <https://doi.org/10.1148/rycan.2020190047>
- Gaur L, Bhandari M, Razdan T, Mallik S, Zhao Z (2022) Explanation-driven deep learning model for prediction of brain Tumour status using MRI image data. *Front Genet* 13:822666. <https://doi.org/10.3389/fgene.2022.822666>
- Ghaffar Nia N, Kaplanoglu E, Nasab A (2023) Evaluation of artificial intelligence techniques in disease diagnosis and prediction. *Discov Artif Intell* 3(1):5. <https://doi.org/10.1007/s44163-023-00005-5>
- Giuste F, Shi W, Zhu Y, Naren T, Isgut M, Sha Y, Tong L, Gupte M, Wang M (2022) Explainable artificial intelligence methods in combating pandemics: a systematic review. *IEEE Rev Biomed Eng* 16:5–21. <https://doi.org/10.48550/arXiv.2112.12705>
- Gómez-Guzmán M, Jiménez L, García-Guerrero E, Lopez-Bonilla O, Tamayo U, Esqueda Elizondo J, Palomino Vizcaino K, Inzunza Gonzalez E (2023) Classifying brain tumors on magnetic resonance imaging by using convolutional neural networks. *Electronics* 12:955. <https://doi.org/10.3390/electronics12040955>
- GTIFA W, Khoja I, SAKLY A (2024) Multimodal MRI brain tumor segmentation using 3D and 3D/2D methods: a study on the MICCAI BRATS dataset. In: 2024 IEEE international conference on advanced systems and emergent technologies (IC_ASET), pp 1–6. IEEE. https://doi.org/10.1109/IC_ASET61847.2024.10596251
- Gundogan E (2025) A novel hybrid deep learning model enhanced with explainable AI for brain tumor multi-classification from MRI images. *Appl Sci* 15(10):5412. <https://doi.org/10.3390/app15105412>
- Gyamfi NK, Goranin N, Ceponis D, Cenys A (2022) Malware detection using convolutional neural network, a deep learning framework: comparative analysis. *J Internet Serv Inf Secur* 12:102–115. <https://doi.org/10.58346/JISIS.2022.14.007>
- Gyamfi NK, Goranin N, Čeponis D, Čenys HA (2022) Malware detection using convolutional neural network, a deep learning framework: comparative analysis. *J Internet Serv Inf Secur* 12(4):102–115. <https://doi.org/10.58346/JISIS.2022.14.007>
- Hakkoum H, Abnane I, Idri A (2022) Interpretability in the medical field: a systematic mapping and review study. *Appl Soft Comput* 117:108391. <https://doi.org/10.1016/j.asoc.2022.108391>
- Hassija V, Chamola V, Mahapatra A, Singal A, Goel D, Huang K, Hussain A (2024) Interpreting black-box models: a review on explainable artificial intelligence. *Cogn Comput* 16(1):45–74. <https://doi.org/10.1007/s12559-023-10158-3>
- He W, Cheng X, Zhao X, Zhou G, Zhu H, Zhao E, Qian G (2023) An interval construction belief rule base with interpretability for complex systems. *Expert Syst Appl* 229(PA). <https://doi.org/10.1016/j.eswa.2023.120485>
- Houssein EH, Gamal AM, Younis EM, Mohamed E (2025) Explainable artificial intelligence for medical imaging systems using deep learning: a comprehensive review. *Clust Comput* 28(7):469. <https://doi.org/10.1007/s10586-025-05281-5>
- Hu C, Xia T, Cui Y, Zou Q, Wang Y-C, Xiao W, Ju S, Li X (2024) Trustworthy multi-phase liver tumor segmentation via evidence-based uncertainty. *Eng Appl Artif Intell* 133:108289. <https://doi.org/10.1016/j.engappai.2024.108289>
- Jacqueline C, Koppaiyan RS (2024) Enriched deep neural network improved by chaotic Harris hawk optimizer for prediction of behavioural traits of individuals. *J Internet Serv Inf Secur* 14:511–523. <https://doi.org/10.58346/JISIS.2024.14.032>
- Jena B, Saxena S, Nayak G, Balestrieri A, Gupta N, Khanna N, Laird J, Kalra M, Fouda M, Saba L, Suri J (2022) Brain tumor characterization using radiogenomics in artificial intelligence framework. *Cancers* 14:4052. <https://doi.org/10.3390/cancers14164052>
- Jun W, Liyuan Z (2022) Brain tumor classification based on attention-guided deep learning model. *Int J Comput Intell Syst* 15(1):35. <https://doi.org/10.2991/ijcis.k.220917.006>
- Kang J, Ullah Z, Gwak J (2021) MRI-based brain tumor classification using an ensemble of deep features and machine learning classifiers. *Sensors* 21(6):2222. <https://doi.org/10.3390/s21062222>
- Khan S, Sajjad M, Hussain T, Ullah A, Imran AS (2020) A review of traditional machine learning and deep learning models for WBCs classification in blood smear images. *IEEE Access* 9:10657–10673. <https://doi.org/10.1109/ACCESS.2020.3047971>

- Khan P, Kader MF, Islam SR, Rahman AB, Kamal MS, Toha MU, Kwak KS (2021) Machine learning and deep learning approaches for brain disease diagnosis: principles and recent advances. *IEEE Access* 9:37622–37655. <https://doi.org/10.1109/ACCESS.2021.3063264>
- Khater T, Hussain A, Bendardaf R, Talaat I, Tawfik H, Ansari S, Mahmoud S (2023) An explainable artificial intelligence model for the classification of breast cancer. *IEEE Access*. <https://doi.org/10.1109/ACCESS.2023.3308446>
- Kodipalli A, Fernandes S, Dasar S (2024) An empirical evaluation of a novel ensemble deep neural network model and explainable AI for accurate segmentation and classification of ovarian tumors using CT images. *Diagnostics* 14:543. <https://doi.org/10.3390/diagnostics14050543>
- Kumar A (2023) Study and analysis of different segmentation methods for brain tumor MRI application. *Multimed Tools Appl* 82(5):7117–7139. <https://doi.org/10.1007/s11042-022-13427-x>
- Kumar Y, Gupta S, Singla R, Hu YC (2022) A systematic review of artificial intelligence techniques in cancer prediction and diagnosis. *Archiv Comput Methods Eng* 29(4):2043–2070. <https://doi.org/10.1007/s11831-021-09652-8>
- Kumar K, Jyoti K (2024) Explainable AI in brain tumor diagnosis: a critical review of ML and DL techniques. Preprint (J Inf Pend)
- Lamba K, Rani S, Shabaz M (2025) Synergizing advanced algorithm of explainable artificial intelligence with hybrid model for enhanced brain tumor detection in healthcare. *Sci Rep* 15(1):20489. <https://doi.org/10.1038/s41598-025-07524-2>
- Lima AA, Mridha MF, Das SC, Kabir MM, Islam MR, Watanobe Y (2022) A comprehensive survey on the detection, classification, and challenges of neurological disorders. *Biology* 11(3):469. <https://doi.org/10.3390/biology11030469>
- Li B, Mao J, Sun Z, Li C, Zhao Q, Tanaka T (2024) Maprotonet: a multi-scale attentive interpretable prototypical part network for 3d magnetic resonance imaging brain tumor classification <https://doi.org/10.48550/arXiv.2404.08917>. arXiv preprint [arXiv:2404.08917](https://arxiv.org/abs/2404.08917)
- Maqsood S, Damaševičius R, Maskeliūnas R (2020) Multi-modal brain tumor detection using deep neural network and multiclass SVM. *Medicina (Kaunas)* 58:1–19. <https://doi.org/10.3390/medicina58081090>
- Marmolejo-Saucedo J, Köse U (2022) Numerical grad-cam based explainable convolutional neural network for brain tumor diagnosis. *Mobile Netw Appl* 29:109–118. <https://doi.org/10.1007/s11036-022-02021-6>
- Mehrotra R, Ansari M, Agrawal R, Anand RS (2020) A transfer learning approach for AI-based classification of brain tumors. *Mach Learn Appl* 2:100003. <https://doi.org/10.1016/j.mlwa.2020.100003>
- Mehta R, Filos A, Baid U, Sako C, McKinley R, Rebsamen M, Dätwyler K, Meier R, Radojewski P, Murugesan GK, Nalawade S, Ganesh C, Wagner B, Yu F, Fei B, Madhuranthakam A, Maldjian J, Daza L, Gómez C, Arbel T (2022) Qu-brats: Miccai brats 2020 challenge on quantifying uncertainty in brain tumor segmentation - analysis of ranking scores and benchmarking results. *J Mach Learn Biomed Imaging*. <https://doi.org/10.59275/j.melba.2022-354b>
- Montaha S, Azam S, Rafid ARH, Hasan MZ, Karim A, Islam A (2022) Timedistributed-CNN-LSTM: a hybrid approach combining CNN and LSTM to classify brain tumors on 3D MRI scans performing ablation study. *IEEE Access* 10:60039–60059. <https://doi.org/10.1109/ACCESS.2022.3181071>
- Murthy MYB, Koteswararao A, Babu MS (2022) Adaptive fuzzy deformable fusion and optimized CNN with ensemble classification for automated brain tumor diagnosis. *Biomed Eng Lett* 12(1):37–58. <https://doi.org/10.1007/s13534-021-00200-4>
- Nahiduzzaman M, Abdulrazak LF, Kibria HB, Khandakar A, Ayari MA, Ahamed MF, Ahsan M, Haider J, Moni MA, Kowalski M (2025) A hybrid explainable model based on advanced machine learning and deep learning models for classifying brain tumors using mri images. *Sci Rep* 15(1):1649. <https://doi.org/10.1038/s41598-025-85874-7>
- Nawaz SA, Khan DM, Qadri S (2022) Brain tumor classification based on hybrid optimized multi-features analysis using magnetic resonance imaging dataset. *Appl Artif Intell* 36(1):2031824. <https://doi.org/10.1080/08839514.2022.2031824>
- Nazir S, Dickson DM, Akram MU (2023) Survey of explainable artificial intelligence techniques for biomedical imaging with deep neural networks. *Comput Biol Med* 156:106668. <https://doi.org/10.1016/j.combiomed.2023.106668>
- Nisar D-E-M, Amin R, Shah N-U-H, Al Ghamdi M, Almotiri S, Alruily M (2021) Healthcare techniques through deep learning: issues, challenges and opportunities. *IEEE Access* PP 1–1. <https://doi.org/10.1109/ACCESS.2021.3095312>
- Raghavendra U, Gudigar A, Paul A, Goutham TS, Inamdar MA, Hegde A, Devi A, Ooi CP, Deo RC, Barua PD, Molinari F, Ciaccio EJ, Acharya UR (2023) Brain tumor detection and screening using artificial intelligence techniques: current trends and future perspectives. *Comput Biol Med* 163:107063. <https://doi.org/10.1016/j.combiomed.2023.107063>
- Ramakrishnananda P, Katiyar Sarmah R (2024) Vu-net: an explainable AI approach for liver segmentation, pp. 1–7. <https://doi.org/10.1109/ICCCNT61001.2024.10725563>

- Rammurthy D, Mahesh PK (2022) Whale Harris hawks optimization-based deep learning classifier for brain tumor detection using MRI images. *J King Saud Univ-Comput Inf Sci* 34(6):3259–3272. <https://doi.org/10.1016/j.jksuci.2021.12.010>
- Ramtekkar PK, Pandey A, Pawar MK (2023) Accurate detection of brain tumor using optimized feature selection based on deep learning techniques. *Multimed Tools Appl* 82(29):44623–44653. <https://doi.org/10.1007/s11042-023-15239-7>
- Rao KN, Khalaf OI, Krishnasree V, Kumar AS, Alsekait DM, Priyanka SS, AbdElminaam DS (2024) An efficient brain tumor detection and classification using pre-trained convolutional neural network models. *Heliyon* 10(17)
- Rasool M, Ismail NA, Boulila W, Ammar A, Samma H, Yafouz W, Emara A-H (2022) A hybrid deep learning model for brain tumour classification. *Entropy* 24:799. <https://doi.org/10.3390/e24060799>
- Raza A, Ayub H, Khan J, Ahmad I, Salama EA, Daradkeh D, Javeed, Rehman A, Hamam H (2022) A hybrid deep learning-based approach for brain tumor classification. *Electronics* 11:1146. <https://doi.org/10.3390/electronics11071146>
- Rehman A, Khan M, Saba T, Mehmood Z, Tariq U, Ayesha N (2020) Microscopic brain tumor detection and classification using 3d CNN and feature selection architecture. *Microsc Res Tech*. <https://doi.org/10.1002/jemt.23597>
- Rosnelly R, Riza B, S S (2023) Comparative analysis of support vector machine and convolutional neural network for malaria parasite classification and feature extraction. *J Wirel Mobile Netw, Ubiquitous Comput Dependable Appl* 14:194–217. <https://doi.org/10.58346/JOWUA.2023.13.015>
- Rustom F, Moroze E, Parva P, Ogmen H, Yazdanbakhsh A (2023) Deep learning and transfer learning for brain tumor detection and classification. *Biol Methods Protoc* 9(1):080. <https://doi.org/10.1093/biomehtods/bpae080>
- Kumar S, Vinod Kumar RS (2024) Literature survey on deep learning methods for liver segmentation from CT images: a comprehensive review. *Multimed Tools Appl* 83:1–30. <https://doi.org/10.1007/s11042-024-18388-5>
- Sadeghi Z, Alizadehsani R, Cifci MA, Kausar S, Rehman R, Mahanta P, Bora K, Almasri A, Alkhawaldeh R, Hussain S, Alatas B, Shoeibi A, Moosaei H, Hladik M, Nahavandi S, Pardalos P, Jammu A, Kashmir K (2024) Kashmir: a review of explainable artificial intelligence in healthcare. *Comput Electric Eng* 118:109370. <https://doi.org/10.1016/j.compeleceng.2024.109370>
- Saeedi S, Rezayi S, Keshavarz H, Kalhori Rostam Niakan S (2023) MRI-based brain tumor detection using convolutional deep learning methods and chosen machine learning techniques. *BMC Med Inform Decis Mak* 23(1):16. <https://doi.org/10.1186/s12911-023-02114-6>
- Saranya A, Subhashini R (2023) A systematic review of explainable artificial intelligence models and applications: recent developments and future trends. *Decis Anal J* 7:100230. <https://doi.org/10.1016/j.dajour.2023.100230>
- Shah M, Shandilya A, Patel K, Mehta M, Sanghavi J, Pandya A (2024) Neuropsychological detection and prediction using machine learning algorithms: a comprehensive review. *Intell Med* 4(3):177–187. <https://doi.org/10.1016/j.imed.2023.12.004>
- Sharif MI, Khan MA, Alhusssein M, Aurangzeb K, Raza M (2021) A decision support system for multimodal brain tumor classification using deep learning. *Complex Intell Syst*. <https://doi.org/10.1007/s40747-021-00508-w>
- Shawon MTR, Shibli GS, Ahmed F, Joy SKS (2025) Explainable cost-sensitive deep neural networks for brain tumor detection from brain MRI images considering data imbalance. *Multimed Tools Appl*. <https://doi.org/10.1007/s11042-025-20842-x>
- Shoaip N, Rezk A, El-Sappagh S, Abuhmed T, Barakat S, Elmogy M (2021) Alzheimer's disease diagnosis based on a semantic rule-based modeling and reasoning approach. *Comput Mater Continua* 69:3531–3548. <https://doi.org/10.32604/cmc.2021.019069>
- Shukla A, Abrol V, Chabra A, Saroch T (2024) Comparative analysis of ML models for brain tumor detection, pp. 193–204. https://doi.org/10.1007/978-981-97-1923-5_15
- Singh A, Sengupta S, Lakshminarayanan V (2020) Explainable deep learning models in medical image analysis. *J Imaging* 6(6):52. <https://doi.org/10.3390/jimaging6060052>
- Sinha A, Rai R, Kumar A, Varma S, Sen S (2023) Explainable-AI based model for brain tumor detection. *IJARCCCE*. <https://doi.org/10.17148/IJARCCCE.2023.12631>
- Solanki S, Singh DU, Chouhan S, Jain S (2023) Brain tumor detection and classification using intelligence techniques: an overview. *IEEE Access* PP, 1–1. <https://doi.org/10.1109/ACCESS.2023.3242666>
- Tjoa E, Guan C (2020) A survey on explainable artificial intelligence (XAI): Toward medical XAI. *IEEE Trans Neural Netw Learn Syst* 32(11):4793–4813. <https://doi.org/10.1109/TNNLS.2020.3020073>
- Tufail AB, Ma YK, Kaabar MK, Martínez F, Funejo AR, Ullah I, Khan R (2021) Deep learning in cancer diagnosis and prognosis prediction: a minireview on challenges, recent trends, and future directions. *Comput Math Methods Med* 2021(1):9025470. <https://doi.org/10.1155/2021/9025470>

- Tummala S, Kadry S, Bukhari SAC, Rauf HT (2022) Classification of brain tumor from magnetic resonance imaging using vision transformers ensembling. *Curr Oncol* 29(10):7498–7511. <https://doi.org/10.3390/curroncol29100607>
- Ullah N, Khan JA, Khan MS, Khan W, Hassan I, Obayya M, Salama AS (2022) An effective approach to detect and identify brain tumors using transfer learning. *Appl Sci* 12(11):5645. <https://doi.org/10.3390/app12115645>
- Vankdothu R, Hameed M (2022) Brain tumor MRI images identification and classification based on the recurrent convolutional neural network. *Measurement Sens* 24:100412. <https://doi.org/10.1016/j.measen.2022.100412>
- Wagan SA, Koo J, Siddiqui IF, Attique M, Shin DR, Qureshi NMF (2022) Internet of medical things and trending converged technologies: a comprehensive review on real-time applications. *J King Saud Univ-Comput Inf Sci* 34(10):9228–9251. <https://doi.org/10.1016/j.jksuci.2022.02.015>
- Yang Y, Zhang Y, Li Y (2023) Artificial intelligence applications in pediatric oncology diagnosis. *Explor Target Anti-tumor Therapy* 4(1):157–169. <https://doi.org/10.37349/etat.2023.00127>
- Yurtsever MME, Atay Y, Arslan B, Sagioglu S (2024) Development of brain tumor radiogenomic classification using GAN-based augmentation of MRI slices in the newly released gazi brains dataset. *BMC Med Inform Decis Mak* 24(1):285. <https://doi.org/10.1186/s12911-024-02699-6>
- Zahoor MM, Qureshi SA, Bibi S, Khan SH, Khan A, Ghafoor U, Bhutta MR (2022) A new deep hybrid boosted and ensemble learning-based brain tumor analysis using MRI. *Sensors* 22(7):2726. <https://doi.org/10.3390/s22072726>
- Zangana H, Mohammed A, Mustafa Alfaqi F (2024) Advancements and applications of convolutional neural networks in image analysis. A comprehensive review. *J Ilm Comput Sci* 3:16–29. <https://doi.org/10.58602/jics.v3i1.30>
- Zeineldin R, Karar M, Elshaer Z, Coburger J, Wirtz R, Burgert O, Mathis-Ullrich F (2022) Explainability of deep neural networks for MRI analysis of brain tumors. *Int J Comput Assist Radiol Surg* 17(9):1673–1683. <https://doi.org/10.1007/s11548-022-02619-x>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Authors and Affiliations

Lincy Annet Abraham¹  · Gopinath Palanisamy¹  · Goutham Veerapu¹  · J. S. Nisha¹ 

✉ Gopinath Palanisamy
gopinath.p@vit.ac.in

Lincy Annet Abraham
lincyannet.abraham2023@vitstudent.ac.in

Goutham Veerapu
veerapu.goutham@vit.ac.in

J. S. Nisha
nisha.js@vit.ac.in

¹ School of Electronics Engineering, Vellore Institute of Technology, Vellore, Tamilnadu 632014, India